

```
1  # apps/api/main.py
2  # -*- coding: utf-8 -*-
3  from __future__ import annotations
4  import os
5  import os
6  from pathlib import Path
7  from fastapi import FastAPI
8  try:
9      from fastapi import FastAPI
10     from fastapi.responses import HTMLResponse
11     from fastapi.staticfiles import StaticFiles
12 except Exception: # pragma: no cover
13     FastAPI = None # type: ignore[assignment]
14 def create_app() -> "FastAPI":
15     from apps.api.routers import health_router, packs_router
16     raise RuntimeError("FastAPI is not installed. Install dependencies
17 before running the API.")
18 APP_NAME = '智能储能电池均衡控制系统'
19     app.include_router(health_router)
20     app.include_router(packs_router)
21 def create_app() -> "FastAPI":
22     if FastAPI is None:
23         raise RuntimeError("FastAPI is not installed. Install dependencies
24 before running the API.")
25     app = FastAPI(title=APP_NAME)
26     return app
27     app.include_router(health_router)
28     app.include_router(packs_router)
29     import uvicorn
30     web_root = Path(__file__).resolve().parents[1] / "web"
31     app.mount("/static", StaticFiles(directory=str(web_root)), name="static")
32 "warning")
33 if __name__ == "__main__":
34     @app.get("/", response_class=HTMLResponse)
35     def index() -> str:
36         return (web_root / "index.html").read_text(encoding="utf-8")
37 from __future__ import annotations
38     return app
39 from .packs import router as packs_router
40 __all__ = ["health_router", "packs_router"]
41 app = create_app()
42 # -*- coding: utf-8 -*-
43 from __future__ import annotations
44 def _main() -> None:
45     import uvicorn
46     @router.get("/health")
47     port = int(os.environ.get("APP_PORT", "8000"))
48     uvicorn.run("apps.api.main:app", host="127.0.0.1", port=port, log_level=
49 "warning")
50 # -*- coding: utf-8 -*-
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51 from __future__ import annotations
52 if __name__ == "__main__":
53     _main()
54 # apps/api/routers/_init__.py
55 # -*- coding: utf-8 -*-
56 from __future__ import annotations
57 def get_store() -> Store:
58     from .health import router as health_router
59     from .packs import router as packs_router
60     db_path = data_dir / "app.db"
61     __all__ = ["health_router", "packs_router"]
62 # apps/api/routers/health.py
63 # -*- coding: utf-8 -*-
64 from __future__ import annotations
65     rows = store.list_packs()
66 from fastapi import APIRouter
67     PackView(
68 router = APIRouter(prefix="/api", tags=["health"])
69     pack_code=r.pack_code,
70     vendor=r.vendor,
71 @router.get("/health")
72 def health() -> dict[str, str]:
73     return {"status": "ok"}
74 # apps/api/routers/packs.py
75 # -*- coding: utf-8 -*-
76 from __future__ import annotations
77     ]
78 import uuid
79 from fastapi import APIRouter, Depends
80 dict[str, str]:
81 from apps.api.schemas.pack import PackCreate, PackView
82 from apps.api.services.store import Store
83     pack_id=pack_id,
84     pack_code=dto.pack_code,
85 router = APIRouter(prefix="/api", tags=["packs"])
86     series_cells=dto.series_cells,
87     parallel_cells=dto.parallel_cells,
88 def get_store() -> Store:
89     project_root = __import__("pathlib").Path(__file__).resolve().parents[3]
90     data_dir = project_root / "data"
91     db_path = data_dir / "app.db"
92     return Store(db_path=db_path)
93 # -*- coding: utf-8 -*-
94 from __future__ import annotations
95 @router.get("/packs", response_model=list[PackView])
96 def list_packs(store: Store = Depends(get_store)) -> list[PackView]:
97     store.seed_demo()
98     rows = store.list_packs()
99     return [
100         PackView(

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101         pack_id=r.pack_id,
102         pack_code=r.pack_code,
103         vendor=r.vendor,
104         series_cells=r.series_cells,
105         parallel_cells=r.parallel_cells,
106         rated_capacity_ah=r.rated_capacity_ah,
107         created_at=r.created_at,
108     )
109     for r in rows
110 ]
111 def utc_now() -> str:
112     return datetime.utcnow().strftime("%Y-%m-%d %H:%M:%S")
113 @router.post("/packs", response_model=dict[str, str])
114 def create_pack(dto: PackCreate, store: Store = Depends(get_store)) ->
115 dict[str, str]:
116     pack_id = f"PACK-{uuid.uuid4().hex[:8].upper()}"
117     store.create_pack(
118         pack_id=pack_id,
119         pack_code=dto.pack_code,
120         vendor=dto.vendor,
121         series_cells=dto.series_cells,
122         parallel_cells=dto.parallel_cells,
123         rated_capacity_ah=dto.rated_capacity_ah,
124     )
125     return {"pack_id": pack_id}
126 # apps/api/domain/__init__.py
127 def _connect(self) -> sqlite3.Connection:
128 # apps/api/domain/enums.py
129 # -*- coding: utf-8 -*-
130 from __future__ import annotations
131 def _init_db(self) -> None:
132 from enum import Enum
133     con.execute(
134         """
135 class BalanceMode(str, Enum):
136     passive = "passive"
137     active = "active"
138     vendor text not null,
139     series_cells integer not null,
140 class AlarmLevel(str, Enum):
141     info = "info"
142     warn = "warn"
143     critical = "critical"
144 # apps/api/services/__init__.py
145 )
146 # apps/api/services/store.py
147 # -*- coding: utf-8 -*-
148 from __future__ import annotations
149     if rows:
150 import sqlite3

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151 from dataclasses import dataclass
152 from datetime import datetime
153 from pathlib import Path
154     vendor="示例能源",
155     series_cells=96,
156 def utc_now() -> str:
157     return datetime.utcnow().strftime("%Y-%m-%d %H:%M:%S")
158 )
159     self.create_pack(
160 @dataclass(frozen=True)
161 class PackRow:
162     pack_id: str
163     pack_code: str
164     vendor: str
165     series_cells: int
166     parallel_cells: int
167     rated_capacity_ah: float
168     created_at: str
169     *,
170     pack_id: str,
171 class Store:
172     def __init__(self, db_path: Path) -> None:
173         self.db_path = db_path
174         self.db_path.parent.mkdir(parents=True, exist_ok=True)
175         self._init_db()
176     ) -> None:
177     def _connect(self) -> sqlite3.Connection:
178         con = sqlite3.connect(str(self.db_path))
179         con.row_factory = sqlite3.Row
180         return con
181 parallel_cells, rated_capacity_ah, created_at)
182     def _init_db(self) -> None:
183         with self._connect() as con:
184             con.execute(
185                 """
186                 create table if not exists packs(
187                     pack_id text primary key,
188                     pack_code text not null,
189                     vendor text not null,
190                     series_cells integer not null,
191                     parallel_cells integer not null,
192                     rated_capacity_ah real not null,
193                     created_at text not null
194                 );
195                 """
196             )
197             con.commit()
198             PackRow(
199 def seed_demo(self) -> None:
200     rows = self.list_packs()
```



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251         return [
252             PackRow(
253                 pack_id=r["pack_id"],
254                 pack_code=r["pack_code"],
255                 vendor=r["vendor"],
256                 series_cells=int(r["series_cells"]),
257                 parallel_cells=int(r["parallel_cells"]),
258                 rated_capacity_ah=float(r["rated_capacity_ah"]),
259                 created_at=str(r["created_at"]),
260             )
261             for r in cur.fetchall()
262         ]
263 # apps/__init__.py
264 <html lang="zh-CN">
265 # apps/api/__init__.py
266 <meta charset="utf-8" />
267 # apps/api/schemas/__init__.py
268 <title>智能储能电池均衡控制系统</title>
269 # apps/api/schemas/pack.py
270 # -*- coding: utf-8 -*-
271 from __future__ import annotations
272     body {margin:0;font-family:Segoe UI,Microsoft YaHei,sans-serif;
273 from pydantic import BaseModel, Field
274     header {position:sticky;top:0;backdrop-filter:blur(10px);
275 background:rgba(17,26,53,.78);border-bottom:1px solid rgba(99,242,210,.18);
276 class PackCreate(BaseModel):
277     pack_code: str = Field(..., description="电池包编号")
278     vendor: str = Field(..., description="厂家")
279     series_cells: int = Field(..., ge=1, le=400, description="串联数量")
280     parallel_cells: int = Field(..., ge=1, le=40, description="并联数量")
281     rated_capacity_ah: float = Field(..., gt=0, description="额定容量(Ah)")
282 grid-template-columns:260px 1fr;gap:16px;}
283     nav {background:rgba(17,26,53,.72);border:1px solid rgba(99,242,210,.12);
284 class PackView(BaseModel):
285     pack_id: str
286     pack_code: str
287     vendor: str
288     series_cells: int
289     parallel_cells: int
290     rated_capacity_ah: float
291     created_at: str
292 # apps/api/tests/__init__.py
293     .grid {display:grid;grid-template-columns:repeat(3,1fr);gap:12px;}
294 # apps/api/tests/test_health.py
295 # -*- coding: utf-8 -*-
296 from __future__ import annotations
297     .hint {color:var(--muted);font-size:12px;line-height:1.6;}
298 import unittest
299     input {padding:10px 12px;border-radius:10px;border:1px solid rgba(99,242,
300 210,.18);background:rgba(255,255,255,.05);color:var(--text);}

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301 class HealthTest(unittest.TestCase):
302     def test_import_app(self) -> None:
303         from apps.api.main import app
304         button.danger{border-color:rgba(255,91,110,.35);background:rgba(255,91,
305             self.assertIsNotNone(app)
306         #modal{position:fixed;inset:0;background:rgba(0,0,0,.55);display:flex;
307     def test_health_endpoint(self) -> None:
308         from fastapi.testclient import TestClient
309         from apps.api.main import app
310         80px rgba(0,0,0,.45);padding:14px;}
311         c = TestClient(app)
312         r = c.get("/api/health")
313         self.assertEqual(r.status_code, 200)
314         self.assertEqual(r.json().get("status"), "ok")
315 </head>
316 <body>
317 if __name__ == "__main__":
318     unittest.main()
319 # apps/web/admin.html
320 <!doctype html>
321 <html lang="zh-CN"><head><meta charset="utf-8" /><meta name="viewport"
322 content="width=device-width, initial-scale=1" />
323 <title>管理员端 - 智能储能电池均衡控制系统</title></head><body style="font-
324 family:Segoe UI,Microsoft YaHei,sans-serif; padding:18px;">
325 <h1>管理员端</h1>
326 <p>面向管理员：参数模板管理、角色权限、审计留痕与策略版本治理。</p>
327 <ol>
328     <li>维护阈值模板（电压差、温升、内阻漂移）。</li>
329     <li>管理用户与角色授权，配置操作留痕策略。</li>
330     <li>发布策略版本并维护回滚点。</li>
331 </ol>
332 </body></html>
333 # apps/web/index.html
334 <!doctype html>
335 <html lang="zh-CN">
336 <head>
337     <meta charset="utf-8" />
338     <meta name="viewport" content="width=device-width, initial-scale=1" />
339     <title>智能储能电池均衡控制系统</title>
340     <style>
341         :root{--bg:#0b1020;--panel:#111a35;--text:#e9eeff;--muted:#a7b2da;--
342         accent:#63f2d2;--warn:#ffb44a;--danger:#ff5b6e;}
343         body{margin:0;font-family:Segoe UI,Microsoft YaHei,sans-serif;
344         background:linear-gradient(180deg,#070a14,var(--bg));color:var(--text);}
345         header{position:sticky;top:0;backdrop-filter:blur(10px);
346         background:rgba(17,26,53,.78);border-bottom:1px solid rgba(99,242,210,.18);
347         padding:14px 18px;display:flex;align-items:center;justify-content:space-
348         between;gap:12px;}
349         header .brand{font-weight:800;letter-spacing:.5px;}
350         header .badge{font-size:12px;color:var(--accent);border:1px solid

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351   rgba(99,242,210,.35);padding:4px 10px;border-radius:999px;}
352   main{max-width:1180px;margin:18px auto;padding:0 18px 28px;display:grid;
353   grid-template-columns:260px 1fr;gap:16px;}
354   nav{background:rgba(17,26,53,.72);border:1px solid rgba(99,242,210,.12);
355   border-radius:14px;padding:12px;}
356   nav a{display:block;color:var(--muted);text-decoration:none;padding:10px
357   10px;border-radius:10px;}
358   nav a:hover{background:rgba(99,242,210,.08);color:var(--text);}
359   .panel{background:rgba(17,26,53,.72);border:1px solid rgba(99,242,210,
360   .12);border-radius:14px;padding:14px;}
361   .k{font-family:ui-monospace,SFMono-Regular,Menlo,Consolas,monospace;
362   color:var(--accent);}
363   .hidden{display:none !important;}
364   .grid{display:grid;grid-template-columns:repeat(3,1fr);gap:12px;}
365   .card{background:rgba(9,12,24,.55);border:1px solid rgba(99,242,210,.12);
366   border-radius:12px;padding:12px;}
367   .card h3{margin:0 0 8px;font-size:14px;color:var(--accent);}
368   .hint{color:var(--muted);font-size:12px;line-height:1.6;}
369   .row{display:grid;gap:10px;margin-top:10px;}
370   input{padding:10px 12px;border-radius:10px;border:1px solid rgba(99,242,
371   210,.18);background:rgba(255,255,255,.05);color:var(--text);}
372   button{cursor:pointer;border-radius:10px;border:1px solid rgba(99,242,
373   210,.22);background:rgba(99,242,210,.10);color:var(--text);padding:8px 12px;}
374   function openModal(title, content){ modalTitle.textContent=title;
375   button:hover{background:rgba(99,242,210,.16);}
376   button.danger{border-color:rgba(255,91,110,.35);background:rgba(255,91,
377   110,.12);}
378   #modal{position:fixed;inset:0;background:rgba(0,0,0,.55);display:flex;
379   align-items:center;justify-content:center;}
380   #modal .dialog{width:min(720px,92vw);background:rgba(17,26,53,.96);
381   border:1px solid rgba(99,242,210,.18);border-radius:14px;box-shadow:0 24px
382   80px rgba(0,0,0,.45);padding:14px;}
383   #modal header{position:static;border:none;background:transparent;
384   padding:0 0 10px;}
385   #modal .title{font-weight:800;}
386   </style>
387   </head>
388   <body>
389   <div id="loginView">
390     <div style="max-width:520px;margin:10vh auto;padding:18px" class="
391     "panel">
392       <h1 style="margin:0 0 8px;font-size:18px;
393       ">智能储能电池均衡控制系统</h1>
394       <div class="hint">V1.0 演示环境：用于截图与材料生成的最小可运行界面。<
395       /div>
396       <form id="login-form" class="row">
397         <label class="hint">用户名 <input name="username" value="admin"
398         autocomplete="username" /></label>
399         <label class="hint">密码 <input name="password" type="password"
400         value="admin123" autocomplete="current-password" /></label>

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401     <button type="submit">登录</button>
402     <div class="hint">提示：任意账号可登录（用于演示与截图）。</div>
403 </form>
404 </div>
405 </div>
406 <div id="appView" style="display:none;">
407 <header>
408     <div class="brand">电池均衡控制台</div>
409     <div class="badge">V1.0</div>
410 </header>
411 <main>
412 <nav>
413     <a href="#home">首页</a>
414     <a href="#packs">电池包</a>
415     <a href="#cells">单体电芯</a>
416     <a href="#balancing">均衡策略</a>
417     <a href="#alarms">告警中心</a>
418     <a href="#reports">统计报表</a>
419     <a href="#audit">审计留痕</a>
420 </nav>
421 <section class="panel">
422     <div style="display:flex;align-items:center;justify-content:space-
423 between;gap:12px;">
424         <h2 id="pageTitle" style="margin:0;font-size:16px;">首页</h2>
425         <div class="hint">API: <span class="k" id=
426 "apiStatus">未检测</span></div>
427     </div>
428     <div id="pageBody" style="margin-top:10px;"></div>
429 </section>
430 </main>
431 </div>
432 <div id="modal" class="modal-shell modal hidden">
433 <div class="dialog" role="dialog" aria-modal="true">
434 <header>
435     <div class="title" id="modalTitle">操作</div>
436     <button class="danger" id="modalClose">关闭</button>
437 </header>
438 <div class="hint" id="modalContent"></div>
439 </div>
440 </div>
441 <script>
442     const modal = document.getElementById('modal');
443     const modalTitle = document.getElementById('modalTitle');
444     const modalContent = document.getElementById('modalContent');
445     function closeModal(){ modal.classList.add('hidden'); }
446     function openModal(title, content){ modalTitle.textContent=title;
447 modalContent.textContent=content; modal.classList.remove('hidden'); }
448     document.getElementById('modalClose').onclick = closeModal;
449     window.__rkModal = { closeModal, openModal };
450     async function apiGet(p){ const r=await fetch(p); if(!r.ok) throw new

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451 Error('HTTP '+r.status); return await r.json(); }
452     function setStatus(ok){ document.getElementById('apiStatus').textContent
453 = ok ? 'OK' : 'FAIL'; }
454     async function ping(){ try{ await apiGet('/api/health'); setStatus(true);
455 } catch(e){ setStatus(false);} }
456     function renderHome(){
457         document.getElementById('pageTitle').textContent='首页';
458         document.getElementById('pageBody').innerHTML = `
459         <div class='grid'>
460             <div class='card'><h3>均衡目标</h3><div class=
461 'hint'>以“最小温升、最小内阻偏差、最短均衡时间”为约束，提升电芯一致性。</div
462 ></div>
463             <div class='card'><h3>在线监测</h3><div class=
464 'hint'>电压/温度/SoC/内阻偏差实时采集，并形成追溯报表。</div></div>
465             <div class='card'><h3>闭环留痕</h3><div class=
466 'hint'>策略下发、执行窗口、告警处置与复核结论自动记录。</div></div>
467         </div>`;
468     }
469     function renderSimple(title, id){
470         document.getElementById('pageTitle').textContent=title;
471         const body = document.getElementById('pageBody');
472         const descMap = {
473             zones: '按站点/柜/簇划分资产边界，提供分区统计、权限隔离与设备定位。',
474             inspections: '对关键页面与接口执行健康检查，形成可追溯的校验记录。',
475             alerts: '对电压差、温升、采样缺失等风险给出提醒，并引导处置闭环。',
476             irrigation: '记录策略下发、执行窗口与回滚点，支撑版本治理与审计。',
477             reports: '输出均衡效果与告警处置报表，支持导出与验收留痕。'
478         };
479         const desc = descMap[id] || '该模块用于展示业务信息与形成材料截图。';
480         body.innerHTML = `<div id='${id}' class='card'><h3>${title}</h3><div
481 class='hint'>${desc}</div><div class='hint'>模块标识: <span class='k'>#${id}
482 </span></div></div>`;
483     }
484 }
485     function renderStations(){
486         document.getElementById('pageTitle').textContent =
487 '工作台（Stations）';
488         const body = document.getElementById('pageBody');
489         body.innerHTML = `
490         <div id="stations" class="card">
491             <h3>Stations 列表</h3>
492             <div class="hint">用于演示“新增/编辑/查看/删除”弹窗操作按钮文案。<
493 /div>
494             <div class="toolbar">
495                 <button onclick="window.__rkModal.openModal('新增',
496 '创建新的均衡任务并填写字段。')">新增</button>
497                 <button onclick="window.__rkModal.openModal('编辑',
498 '修改任务阈值与执行窗口。')">编辑</button>
499                 <button onclick="window.__rkModal.openModal('查看',
500 '查看任务详情与引用关系。')">查看</button>

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501         <button class="danger" onclick="window.__rkModal.openModal('删除
502         ','删除条目前请确认无引用并记录审计。')">删除</button>
503     </div>
504     <div class="hint">提示：请按业务流程依次执行新增、编辑、查看、删除
505     ，并复核弹窗信息完整性。</div>
506     </div>;
507 }
508 function route(){
509     const hash=(location.hash||'#home').replace(/^#/, "");
510     if(hash==='home'||hash==='') return renderHome();
511     if(hash==='zones') return renderSimple('资源分区','zones');
512     if(hash==='stations') return renderStations();
513     if(hash==='inspections') return renderSimple('交互检查','inspections');
514     def index() -> str:
515         if(hash==='alerts') return renderSimple('风险提示','alerts');
516         if(hash==='irrigation') return renderSimple('发布流水','irrigation');
517         if(hash==='reports') return renderSimple('统计报表','reports');
518         return renderSimple('页面', hash);
519     }
520     document.getElementById('login-form').addEventListener('submit', (e)=>{
521         e.preventDefault();
522         localStorage.setItem('__demo_logged_in','1');
523         document.getElementById('loginView').style.display='none';
524         document.getElementById('appView').style.display='block';
525         ping(); route();
526     });
527     window.addEventListener('hashchange', ()=>route());
528     if(localStorage.getItem('__demo_logged_in')==='1'){
529         document.getElementById('loginView').style.display='none';
530         document.getElementById('appView').style.display='block';
531         ping(); route();
532     } else { renderHome(); }
533 </script>
534 </body>
535 </html>
536 # apps/web/student.html
537 <!doctype html>
538 <html lang="zh-CN"><head><meta charset="utf-8" /><meta name="viewport"
539 content="width=device-width, initial-scale=1" />
540 <title>值班员端 - 智能储能电池均衡控制系统</title></head><body style="font-
541 family:Segoe UI,Microsoft YaHei,sans-serif; padding:18px;">
542 <h1>值班员端</h1>
543 <p>面向值班员：告警确认、策略下发、执行窗口控制与风险提示。</p>
544 <ol>
545     <li>确认告警并选择处置路径（观察/降载/停用）。</li>
546     <li>查看均衡策略建议与预计温升。</li>
547     <li>一键生成处置报告并提交复核。</li>
548 </ol>
549 </body></html>
550 # apps/web/teacher.html

```

```

551 <!doctype html>
552 <html lang="zh-CN"><head><meta charset="utf-8" /><meta name="viewport"
553 content="width=device-width, initial-scale=1" />
554 <title>运行维护端 - 智能储能电池均衡控制系统</title></head><body style="font-
555 family:Segoe UI,Microsoft YaHei,sans-serif; padding:18px;">
556 <h1>运行维护端</h1>
557 <p>面向站端运维：设备巡检、数据采样、阈值校准与工单闭环。</p>
558 <ol>
559   <li>查看采样窗口、采样频率与数据质量。</li>
560   <li>对异常电芯做复核采样，并生成校准记录。</li>
561   <li>导出运维日报与告警处置清单。</li>
562 </ol>
563 </body></html>
564 # apps/api/main.py
565 # -*- coding: utf-8 -*-
566 from __future__ import annotations
567 import os
568 from pathlib import Path
569 try:
570     from fastapi import FastAPI
571     from fastapi.responses import HTMLResponse
572     from fastapi.staticfiles import StaticFiles
573 except Exception: # pragma: no cover
574     FastAPI = None # type: ignore[assignment]
575 from apps.api.routers import health_router, packs_router
576 APP_NAME = '智能储能电池均衡控制系统'
577 def create_app() -> "FastAPI":
578     if FastAPI is None:
579         raise RuntimeError("FastAPI is not installed. Install dependencies
580 before running the API.")
581     app = FastAPI(title=APP_NAME)
582     app.include_router(health_router)
583     app.include_router(packs_router)
584     web_root = Path(__file__).resolve().parents[1] / "web"
585     app.mount("/static", StaticFiles(directory=str(web_root)), name="static")
586     @app.get("/", response_class=HTMLResponse)
587     def index() -> str:
588         return (web_root / "index.html").read_text(encoding="utf-8")
589     return app
590 app = create_app()
591 def _main() -> None:
592     import uvicorn
593     port = int(os.environ.get("APP_PORT", "8000"))
594     uvicorn.run("apps.api.main:app", host="127.0.0.1", port=port, log_level=
595 "warning")
596 if __name__ == "__main__":
597     _main()
598 # apps/api/routers/__init__.py
599 # -*- coding: utf-8 -*-
600 from __future__ import annotations

```

```

601 from .health import router as health_router
602 from .packs import router as packs_router
603 __all__ = ["health_router", "packs_router"]
604 # apps/api/routers/health.py
605 # -*- coding: utf-8 -*-
606 from __future__ import annotations
607 from fastapi import APIRouter
608 router = APIRouter(prefix="/api", tags=["health"])
609 @router.get("/health")
610 def health() -> dict[str, str]:
611     return {"status": "ok"}
612 # apps/api/routers/packs.py
613 # -*- coding: utf-8 -*-
614 from __future__ import annotations
615 import uuid
616 from fastapi import APIRouter, Depends
617 from apps.api.schemas.pack import PackCreate, PackView
618 from apps.api.services.store import Store
619 router = APIRouter(prefix="/api", tags=["packs"])
620 def get_store() -> Store:
621     project_root = __import__("pathlib").Path(__file__).resolve().parents[3]
622     data_dir = project_root / "data"
623     db_path = data_dir / "app.db"
624     return Store(db_path=db_path)
625 @router.get("/packs", response_model=list[PackView])
626 def list_packs(store: Store = Depends(get_store)) -> list[PackView]:
627     store.seed_demo()
628     rows = store.list_packs()
629     return [
630         PackView(
631             pack_id=r.pack_id,
632             pack_code=r.pack_code,
633             vendor=r.vendor,
634             series_cells=r.series_cells,
635             parallel_cells=r.parallel_cells,
636             rated_capacity_ah=r.rated_capacity_ah,
637             created_at=r.created_at,
638         )
639         for r in rows
640     ]
641 @router.post("/packs", response_model=dict[str, str])
642 def create_pack(dto: PackCreate, store: Store = Depends(get_store)) ->
643 dict[str, str]:
644     pack_id = f"PACK-{uuid.uuid4().hex[:8].upper()}"
645     store.create_pack(
646         pack_id=pack_id,
647         pack_code=dto.pack_code,
648         vendor=dto.vendor,
649         series_cells=dto.series_cells,
650         parallel_cells=dto.parallel_cells,

```

```

651         rated_capacity_ah=dto.rated_capacity_ah,
652     )
653     return {"pack_id": pack_id}
654 # apps/api/domain/__init__.py
655 # apps/api/domain/enums.py
656 # -*- coding: utf-8 -*-
657 from __future__ import annotations
658 from enum import Enum
659 class BalanceMode(str, Enum):
660     passive = "passive"
661     active = "active"
662 class AlarmLevel(str, Enum):
663     info = "info"
664     warn = "warn"
665     critical = "critical"
666 # apps/api/services/__init__.py
667 # apps/api/services/store.py
668 # -*- coding: utf-8 -*-
669 from __future__ import annotations
670 import sqlite3
671 from dataclasses import dataclass
672 from datetime import datetime
673 from pathlib import Path
674 def utc_now() -> str:
675     return datetime.utcnow().strftime("%Y-%m-%d %H:%M:%S")
676 @dataclass(frozen=True)
677 class PackRow:
678     pack_id: str
679     pack_code: str
680     vendor: str
681     series_cells: int
682     parallel_cells: int
683     rated_capacity_ah: float
684     created_at: str
685 class Store:
686     def __init__(self, db_path: Path) -> None:
687         self.db_path = db_path
688         self.db_path.parent.mkdir(parents=True, exist_ok=True)
689         self._init_db()
690     def _connect(self) -> sqlite3.Connection:
691         con = sqlite3.connect(str(self.db_path))
692         con.row_factory = sqlite3.Row
693         return con
694     def _init_db(self) -> None:
695         with self._connect() as con:
696             con.execute(
697                 """
698                 create table if not exists packs(
699                     pack_id text primary key,
700                     pack_code text not null,

```

```

701         vendor text not null,
702         series_cells integer not null,
703         parallel_cells integer not null,
704         rated_capacity_ah real not null,
705         created_at text not null
706     );
707     """
708 )
709 con.commit()
710 def seed_demo(self) -> None:
711     rows = self.list_packs()
712     if rows:
713         return
714     self.create_pack(
715         pack_id="PACK-0001",
716         pack_code="SZ-ESS-0001",
717         vendor="示例能源",
718         series_cells=96,
719         parallel_cells=1,
720         rated_capacity_ah=280.0,
721     )
722     self.create_pack(
723         pack_id="PACK-0002",
724         pack_code="SZ-ESS-0002",
725         vendor="示例能源",
726         series_cells=96,
727         parallel_cells=1,
728         rated_capacity_ah=280.0,
729     )
730 def create_pack(
731     self,
732     *,
733     pack_id: str,
734     pack_code: str,
735     vendor: str,
736     series_cells: int,
737     parallel_cells: int,
738     rated_capacity_ah: float,
739 ) -> None:
740     with self._connect() as con:
741         con.execute(
742             """
743             insert into packs(pack_id, pack_code, vendor, series_cells,
744             parallel_cells, rated_capacity_ah, created_at)
745             values(?,?,?,?,?,?,?)
746             """,
747             (pack_id, pack_code, vendor, int(series_cells),
748             int(parallel_cells), float(rated_capacity_ah), utc_now()),
749         )
750         con.commit()

```

```

751     def list_packs(self) -> list[PackRow]:
752         with self._connect() as con:
753             cur = con.execute(
754                 """
755                 select pack_id, pack_code, vendor, series_cells,
756 parallel_cells, rated_capacity_ah, created_at
757                 from packs order by created_at desc
758                 """
759             )
760             return [
761                 PackRow(
762                     pack_id=r["pack_id"],
763                     pack_code=r["pack_code"],
764                     vendor=r["vendor"],
765                     series_cells=int(r["series_cells"]),
766                     parallel_cells=int(r["parallel_cells"]),
767                     rated_capacity_ah=float(r["rated_capacity_ah"]),
768                     created_at=str(r["created_at"]),
769                 )
770                 for r in cur.fetchall()
771             ]
772 # apps/__init__.py
773 # apps/api/__init__.py
774 # apps/api/schemas/__init__.py
775 # apps/api/schemas/pack.py
776 # -*- coding: utf-8 -*-
777 from __future__ import annotations
778 from pydantic import BaseModel, Field
779 class PackCreate(BaseModel):
780     pack_code: str = Field(..., description="电池包编号")
781     vendor: str = Field(..., description="厂家")
782     series_cells: int = Field(..., ge=1, le=400, description="串联数量")
783     parallel_cells: int = Field(..., ge=1, le=40, description="并联数量")
784     rated_capacity_ah: float = Field(..., gt=0, description="额定容量(Ah)")
785 class PackView(BaseModel):
786     pack_id: str
787     pack_code: str
788     vendor: str
789     series_cells: int
790     parallel_cells: int
791     rated_capacity_ah: float
792     created_at: str
793 # apps/api/tests/__init__.py
794 # apps/api/tests/test_health.py
795 # -*- coding: utf-8 -*-
796 from __future__ import annotations
797 import unittest
798 class HealthTest(unittest.TestCase):
799     def test_import_app(self) -> None:
800         from apps.api.main import app

```



```

801         self.assertIsNotNone(app)
802     def test_health_endpoint(self) -> None:
803         from fastapi.testclient import TestClient
804         from apps.api.main import app
805         c = TestClient(app)
806         r = c.get("/api/health")
807         self.assertEqual(r.status_code, 200)
808         self.assertEqual(r.json().get("status"), "ok")
809 if __name__ == "__main__":
810     unittest.main()
811 # apps/web/admin.html
812 <!doctype html>
813 <html lang="zh-CN"><head><meta charset="utf-8" /><meta name="viewport"
814 content="width=device-width, initial-scale=1" />
815 <title>管理员端 - 智能储能电池均衡控制系统</title></head><body style="font-
816 family:Segoe UI,Microsoft YaHei,sans-serif; padding:18px;">
817 <h1>管理员端</h1>
818 <p>面向管理员：参数模板管理、角色权限、审计留痕与策略版本治理。</p>
819 <ol>
820     <li>维护阈值模板（电压差、温升、内阻漂移）。</li>
821     <li>管理用户与角色授权，配置操作留痕策略。</li>
822     <li>发布策略版本并维护回滚点。</li>
823 </ol>
824 </body></html>
825 # apps/web/index.html
826 <!doctype html>
827 <html lang="zh-CN">
828 <head>
829     <meta charset="utf-8" />
830     <meta name="viewport" content="width=device-width, initial-scale=1" />
831     <title>智能储能电池均衡控制系统</title>
832     <style>
833         :root{--bg:#0b1020;--panel:#111a35;--text:#e9eef;--muted:#a7b2da;--
834 accent:#63f2d2;--warn:#ffb44a;--danger:#ff5b6e;}
835         body{margin:0;font-family:Segoe UI,Microsoft YaHei,sans-serif;
836 background:linear-gradient(180deg,#070a14,var(--bg));color:var(--text);}
837         header{position:sticky;top:0;backdrop-filter:blur(10px);
838 background:rgba(17,26,53,.78);border-bottom:1px solid rgba(99,242,210,.18);
839 padding:14px 18px;display:flex;align-items:center;justify-content:space-
840 between;gap:12px;}
841         header .brand{font-weight:800;letter-spacing:.5px;}
842         header .badge{font-size:12px;color:var(--accent);border:1px solid
843 rgba(99,242,210,.35);padding:4px 10px;border-radius:999px;}
844         main{max-width:1180px;margin:18px auto;padding:0 18px 28px;display:grid;
845 grid-template-columns:260px 1fr;gap:16px;}
846         nav{background:rgba(17,26,53,.72);border:1px solid rgba(99,242,210,.12);
847 border-radius:14px;padding:12px;}
848         nav a{display:block;color:var(--muted);text-decoration:none;padding:10px
849 10px;border-radius:10px;}
850         nav a:hover{background:rgba(99,242,210,.08);color:var(--text);}

```

```

851     .panel{background:rgba(17,26,53,.72);border:1px solid rgba(99,242,210,
852     .12);border-radius:14px;padding:14px;}
853     .k{font-family:ui-monospace,SFMono-Regular,Menlo,Consolas,monospace;
854     color:var(--accent);}
855     .hidden{display:none !important;}
856     .grid{display:grid;grid-template-columns:repeat(3,1fr);gap:12px;}
857     .card{background:rgba(9,12,24,.55);border:1px solid rgba(99,242,210,.12);
858     border-radius:12px;padding:12px;}
859     .card h3{margin:0 0 8px;font-size:14px;color:var(--accent);}
860     .hint{color:var(--muted);font-size:12px;line-height:1.6;}
861     .row{display:grid;gap:10px;margin-top:10px;}
862     input{padding:10px 12px;border-radius:10px;border:1px solid rgba(99,242,
863     210,.18);background:rgba(255,255,255,.05);color:var(--text);}
864     button{cursor:pointer;border-radius:10px;border:1px solid rgba(99,242,
865     210,.22);background:rgba(99,242,210,.10);color:var(--text);padding:8px 12px;}
866     button:hover{background:rgba(99,242,210,.16);}
867     button.danger{border-color:rgba(255,91,110,.35);background:rgba(255,91,
868     110,.12);}
869     #modal{position:fixed;inset:0;background:rgba(0,0,0,.55);display:flex;
870     align-items:center;justify-content:center;}
871     #modal .dialog{width:min(720px,92vw);background:rgba(17,26,53,.96);
872     border:1px solid rgba(99,242,210,.18);border-radius:14px;box-shadow:0 24px
873     80px rgba(0,0,0,.45);padding:14px;}
874     #modal header{position:static;border:none;background:transparent;
875     padding:0 0 10px;}
876     #modal .title{font-weight:800;}
877     </style>
878     </head>
879     <body>
880     <div id="loginView">
881     <div style="max-width:520px;margin:10vh auto;padding:18px" class=
882     "panel">
883     <h1 style="margin:0 0 8px;font-size:18px;
884     ">智能储能电池均衡控制系统</h1>
885     <div class="hint">V1.0 演示环境：用于截图与材料生成的最小可运行界面。<
886     /div>
887     <form id="login-form" class="row">
888     <label class="hint">用户名 <input name="username" value="admin"
889     autocomplete="username" /></label>
890     <label class="hint">密码 <input name="password" type="password"
891     value="admin123" autocomplete="current-password" /></label>
892     <button type="submit">登录</button>
893     <div class="hint">提示：任意账号可登录（用于演示与截图）。</div>
894     </form>
895     </div>
896     </div>
897     <div id="appView" style="display:none;">
898     <header>
899     <div class="brand">电池均衡控制台</div>
900     <div class="badge">V1.0</div>

```

```

901     </header>
902     <main>
903         <nav>
904             <a href="#home">首页</a>
905             <a href="#packs">电池包</a>
906             <a href="#cells">单体电芯</a>
907             <a href="#balancing">均衡策略</a>
908             <a href="#alarms">告警中心</a>
909             <a href="#reports">统计报表</a>
910             <a href="#audit">审计留痕</a>
911         </nav>
912         <section class="panel">
913             <div style="display:flex;align-items:center;justify-content:space-
914 between;gap:12px;">
915                 <h2 id="pageTitle" style="margin:0;font-size:16px;">首页</h2>
916                 <div class="hint">API: <span class="k" id=
917 "apiStatus">未检测</span></div>
918             </div>
919             <div id="pageBody" style="margin-top:10px;"></div>
920         </section>
921     </main>
922 </div>
923 <div id="modal" class="modal-shell modal hidden">
924     <div class="dialog" role="dialog" aria-modal="true">
925         <header>
926             <div class="title" id="modalTitle">操作</div>
927             <button class="danger" id="modalClose">关闭</button>
928         </header>
929         <div class="hint" id="modalContent"></div>
930     </div>
931 </div>
932 <script>
933     const modal = document.getElementById('modal');
934     const modalTitle = document.getElementById('modalTitle');
935     const modalContent = document.getElementById('modalContent');
936     function closeModal(){ modal.classList.add('hidden'); }
937     function openModal(title, content){ modalTitle.textContent=title;
938 modalContent.textContent=content; modal.classList.remove('hidden'); }
939     document.getElementById('modalClose').onclick = closeModal;
940     window.__rkModal = { closeModal, openModal };
941     async function apiGet(p){ const r=await fetch(p); if(!r.ok) throw new
942 Error('HTTP '+r.status); return await r.json(); }
943     function setStatus(ok){ document.getElementById('apiStatus').textContent
944 = ok ? 'OK' : 'FAIL'; }
945     async function ping(){ try{ await apiGet('/api/health'); setStatus(true);
946 } catch(e){ setStatus(false);} }
947     function renderHome(){
948         document.getElementById('pageTitle').textContent='首页';
949         document.getElementById('pageBody').innerHTML = `
950         <div class='grid'>

```

```

951         <div class='card'><h3>均衡目标</h3><div class=
952 'hint'>以“最小温升、最小内阻偏差、最短均衡时间”为约束，提升电芯一致性。</div
953 ></div>
954         <div class='card'><h3>在线监测</h3><div class=
955 'hint'>电压/温度/SoC/内阻偏差实时采集，并形成追溯报表。</div></div>
956         <div class='card'><h3>闭环留痕</h3><div class=
957 'hint'>策略下发、执行窗口、告警处置与复核结论自动记录。</div></div>
958         </div>';
959     }
960     function renderSimple(title, id){
961         document.getElementById('pageTitle').textContent=title;
962         const body = document.getElementById('pageBody');
963         const descMap = {
964             zones: '按站点/柜/簇划分资产边界，提供分区统计、权限隔离与设备定位。
965 ',
966             inspections: '对关键页面与接口执行健康检查，形成可追溯的校验记录。',
967             alerts: '对电压差、温升、采样缺失等风险给出提醒，并引导处置闭环。',
968             irrigation: '记录策略下发、执行窗口与回滚点，支撑版本治理与审计。',
969             reports: '输出均衡效果与告警处置报表，支持导出与验收留痕。'
970         };
971         const desc = descMap[id] || '该模块用于展示业务信息与形成材料截图。';
972         body.innerHTML = `<div id='${id}' class='card'><h3>${title}</h3><div
973 class='hint'>${desc}</div><div class='hint'>模块标识：<span class='k'>#${id}
974 </span></div></div>`;
975     }
976     function renderStations(){
977         document.getElementById('pageTitle').textContent =
978 '工作台（Stations）';
979         const body = document.getElementById('pageBody');
980         body.innerHTML = `
981         <div id="stations" class="card">
982             <h3>Stations 列表</h3>
983             <div class="hint">用于演示“新增/编辑/查看/删除”弹窗操作按钮文案。<
984 /div>
985             <div class="toolbar">
986                 <button onclick="window.__rkModal.openModal('新增',
987 '创建新的均衡任务并填写字段。')">新增</button>
988                 <button onclick="window.__rkModal.openModal('编辑',
989 '修改任务阈值与执行窗口。')">编辑</button>
990                 <button onclick="window.__rkModal.openModal('查看',
991 '查看任务详情与引用关系。')">查看</button>
992                 <button class="danger" onclick="window.__rkModal.openModal('删除
993 ';删除条目前请确认无引用并记录审计。')">删除</button>
994             </div>
995             <div class="hint">提示：请按业务流程依次执行新增、编辑、查看、删除
996 ，并复核弹窗信息完整性。</div>
997             </div>`;
998     }
999     function route(){
1000         const hash=(location.hash||'#home').replace(/^#/, "");

```

```
1001     if(hash==='home'||hash==='') return renderHome();
1002     if(hash==='zones') return renderSimple('资源分区','zones');
1003     if(hash==='stations') return renderStations();
1004     if(hash==='inspections') return renderSimple('交互检查','inspections');
1005     if(hash==='alerts') return renderSimple('风险提示','alerts');
1006     if(hash==='irrigation') return renderSimple('发布流水','irrigation');
1007     if(hash==='reports') return renderSimple('统计报表','reports');
1008     return renderSimple('页面', hash);
1009 }
1010 document.getElementById('login-form').addEventListener('submit', (e)=>{
1011     e.preventDefault();
1012     localStorage.setItem('__demo_logged_in','1');
1013     document.getElementById('loginView').style.display='none';
1014     document.getElementById('appView').style.display='block';
1015     ping(); route();
1016 });
1017 window.addEventListener('hashchange', ()=>route());
1018 if(localStorage.getItem('__demo_logged_in')==='1'){
1019     document.getElementById('loginView').style.display='none';
1020     document.getElementById('appView').style.display='block';
1021     ping(); route();
1022 } else { renderHome(); }
1023 </script>
1024 </body>
1025 </html>
1026 # apps/web/student.html
1027 <!doctype html>
1028 <html lang="zh-CN"><head><meta charset="utf-8" /><meta name="viewport"
1029 content="width=device-width, initial-scale=1" />
1030 <title>值班员端 - 智能储能电池均衡控制系统</title></head><body style="font-
1031 family:Segoe UI,Microsoft YaHei,sans-serif; padding:18px;">
1032 <h1>值班员端</h1>
1033 <p>面向值班员：告警确认、策略下发、执行窗口控制与风险提示。</p>
1034 <ol>
1035     <li>确认告警并选择处置路径（观察/降载/停用）。</li>
1036     <li>查看均衡策略建议与预计温升。</li>
1037     <li>一键生成处置报告并提交复核。</li>
1038 </ol>
1039 </body></html>
1040 # apps/web/teacher.html
1041 <!doctype html>
1042 <html lang="zh-CN"><head><meta charset="utf-8" /><meta name="viewport"
1043 content="width=device-width, initial-scale=1" />
1044 <title>运行维护端 - 智能储能电池均衡控制系统</title></head><body style="font-
1045 family:Segoe UI,Microsoft YaHei,sans-serif; padding:18px;">
1046 <h1>运行维护端</h1>
1047 <p>面向站端运维：设备巡检、数据采样、阈值校准与工单闭环。</p>
1048 <ol>
1049     <li>查看采样窗口、采样频率与数据质量。</li>
1050     <li>对异常电芯做复核采样，并生成校准记录。</li>
```

```

1051     <li>导出运维日报与告警处置清单。</li>
1052 </ol>
1053 </body></html>
1054 # apps/api/main.py
1055 # -*- coding: utf-8 -*-
1056 from __future__ import annotations
1057 import os
1058 from pathlib import Path
1059 try:
1060     from fastapi import FastAPI
1061     from fastapi.responses import HTMLResponse
1062     from fastapi.staticfiles import StaticFiles
1063 except Exception: # pragma: no cover
1064     FastAPI = None # type: ignore[assignment]
1065 from apps.api.routers import health_router, packs_router
1066 APP_NAME = '智能储能电池均衡控制系统'
1067 def create_app() -> "FastAPI":
1068     if FastAPI is None:
1069         raise RuntimeError("FastAPI is not installed. Install dependencies
1070 before running the API.")
1071     app = FastAPI(title=APP_NAME)
1072     app.include_router(health_router)
1073     app.include_router(packs_router)
1074     web_root = Path(__file__).resolve().parents[1] / "web"
1075     app.mount("/static", StaticFiles(directory=str(web_root)), name="static")
1076     @app.get("/", response_class=HTMLResponse)
1077     def index() -> str:
1078         return (web_root / "index.html").read_text(encoding="utf-8")
1079     return app
1080 app = create_app()
1081 def _main() -> None:
1082     import uvicorn
1083     port = int(os.environ.get("APP_PORT", "8000"))
1084     uvicorn.run("apps.api.main:app", host="127.0.0.1", port=port, log_level=
1085 "warning")
1086 if __name__ == "__main__":
1087     _main()
1088 # apps/api/routers/_init_.py
1089 # -*- coding: utf-8 -*-
1090 from __future__ import annotations
1091 from .health import router as health_router
1092 from .packs import router as packs_router
1093 __all__ = ["health_router", "packs_router"]
1094 # apps/api/routers/health.py
1095 # -*- coding: utf-8 -*-
1096 from __future__ import annotations
1097 from fastapi import APIRouter
1098 router = APIRouter(prefix="/api", tags=["health"])
1099 @router.get("/health")
1100 def health() -> dict[str, str]:

```

```

1101     return {"status": "ok"}
1102 # apps/api/routers/packs.py
1103 # -*- coding: utf-8 -*-
1104 from __future__ import annotations
1105 import uuid
1106 from fastapi import APIRouter, Depends
1107 from apps.api.schemas.pack import PackCreate, PackView
1108 from apps.api.services.store import Store
1109 router = APIRouter(prefix="/api", tags=["packs"])
1110 def get_store() -> Store:
1111     project_root = __import__("pathlib").Path(__file__).resolve().parents[3]
1112     data_dir = project_root / "data"
1113     db_path = data_dir / "app.db"
1114     return Store(db_path=db_path)
1115 @router.get("/packs", response_model=list[PackView])
1116 def list_packs(store: Store = Depends(get_store)) -> list[PackView]:
1117     store.seed_demo()
1118     rows = store.list_packs()
1119     return [
1120         PackView(
1121             pack_id=r.pack_id,
1122             pack_code=r.pack_code,
1123             vendor=r.vendor,
1124             series_cells=r.series_cells,
1125             parallel_cells=r.parallel_cells,
1126             rated_capacity_ah=r.rated_capacity_ah,
1127             created_at=r.created_at,
1128         )
1129         for r in rows
1130     ]
1131 @router.post("/packs", response_model=dict[str, str])
1132 def create_pack(dto: PackCreate, store: Store = Depends(get_store)) ->
1133 dict[str, str]:
1134     pack_id = f"PACK-{uuid.uuid4().hex[:8].upper()}"
1135     store.create_pack(
1136         pack_id=pack_id,
1137         pack_code=dto.pack_code,
1138         vendor=dto.vendor,
1139         series_cells=dto.series_cells,
1140         parallel_cells=dto.parallel_cells,
1141         rated_capacity_ah=dto.rated_capacity_ah,
1142     )
1143     return {"pack_id": pack_id}
1144 # apps/api/domain/__init__.py
1145 # apps/api/domain/enums.py
1146 # -*- coding: utf-8 -*-
1147 from __future__ import annotations
1148 from enum import Enum
1149 class BalanceMode(str, Enum):
1150     passive = "passive"

```

```
1151     active = "active"
1152 class AlarmLevel(str, Enum):
1153     info = "info"
1154     warn = "warn"
1155     critical = "critical"
1156 # apps/api/services/__init__.py
1157 # apps/api/services/store.py
1158 # -*- coding: utf-8 -*-
1159 from __future__ import annotations
1160 import sqlite3
1161 from dataclasses import dataclass
1162 from datetime import datetime
1163 from pathlib import Path
1164 def utc_now() -> str:
1165     return datetime.utcnow().strftime("%Y-%m-%d %H:%M:%S")
1166 @dataclass(frozen=True)
1167 class PackRow:
1168     pack_id: str
1169     pack_code: str
1170     vendor: str
1171     series_cells: int
1172     parallel_cells: int
1173     rated_capacity_ah: float
1174     created_at: str
1175 class Store:
1176     def __init__(self, db_path: Path) -> None:
1177         self.db_path = db_path
1178         self.db_path.parent.mkdir(parents=True, exist_ok=True)
1179         self._init_db()
1180     def _connect(self) -> sqlite3.Connection:
1181         con = sqlite3.connect(str(self.db_path))
1182         con.row_factory = sqlite3.Row
1183         return con
1184     def _init_db(self) -> None:
1185         with self._connect() as con:
1186             con.execute(
1187                 """
1188                 create table if not exists packs(
1189                     pack_id text primary key,
1190                     pack_code text not null,
1191                     vendor text not null,
1192                     series_cells integer not null,
1193                     parallel_cells integer not null,
1194                     rated_capacity_ah real not null,
1195                     created_at text not null
1196                 );
1197                 """
1198             )
1199             con.commit()
1200     def seed_demo(self) -> None:
```



```

1201         rows = self.list_packs()
1202         if rows:
1203             return
1204         self.create_pack(
1205             pack_id="PACK-0001",
1206             pack_code="SZ-ESS-0001",
1207             vendor="示例能源",
1208             series_cells=96,
1209             parallel_cells=1,
1210             rated_capacity_ah=280.0,
1211         )
1212         self.create_pack(
1213             pack_id="PACK-0002",
1214             pack_code="SZ-ESS-0002",
1215             vendor="示例能源",
1216             series_cells=96,
1217             parallel_cells=1,
1218             rated_capacity_ah=280.0,
1219         )
1220     def create_pack(
1221         self,
1222         *,
1223         pack_id: str,
1224         pack_code: str,
1225         vendor: str,
1226         series_cells: int,
1227         parallel_cells: int,
1228         rated_capacity_ah: float,
1229     ) -> None:
1230         with self._connect() as con:
1231             con.execute(
1232                 """
1233                 insert into packs(pack_id, pack_code, vendor, series_cells,
1234 parallel_cells, rated_capacity_ah, created_at)
1235                 values(?,?,?,?,?,?,?)
1236                 """,
1237                 (pack_id, pack_code, vendor, int(series_cells),
1238 int(parallel_cells), float(rated_capacity_ah), utc_now()),
1239             )
1240             con.commit()
1241     def list_packs(self) -> list[PackRow]:
1242         with self._connect() as con:
1243             cur = con.execute(
1244                 """
1245                 select pack_id, pack_code, vendor, series_cells,
1246 parallel_cells, rated_capacity_ah, created_at
1247                 from packs order by created_at desc
1248                 """,
1249             )
1250             return [

```

```

1251         PackRow(
1252             pack_id=r["pack_id"],
1253             pack_code=r["pack_code"],
1254             vendor=r["vendor"],
1255             series_cells=int(r["series_cells"]),
1256             parallel_cells=int(r["parallel_cells"]),
1257             rated_capacity_ah=float(r["rated_capacity_ah"]),
1258             created_at=str(r["created_at"]),
1259         )
1260     for r in cur.fetchall()
1261 ]
1262 # apps/__init__.py
1263 # apps/api/__init__.py
1264 # apps/api/schemas/__init__.py
1265 # apps/api/schemas/pack.py
1266 # -*- coding: utf-8 -*-
1267 from __future__ import annotations
1268 from pydantic import BaseModel, Field
1269 class PackCreate(BaseModel):
1270     pack_code: str = Field(..., description="电池包编号")
1271     vendor: str = Field(..., description="厂家")
1272     series_cells: int = Field(..., ge=1, le=400, description="串联数量")
1273     parallel_cells: int = Field(..., ge=1, le=40, description="并联数量")
1274     rated_capacity_ah: float = Field(..., gt=0, description="额定容量(Ah)")
1275 class PackView(BaseModel):
1276     pack_id: str
1277     pack_code: str
1278     vendor: str
1279     series_cells: int
1280     parallel_cells: int
1281     rated_capacity_ah: float
1282     created_at: str
1283 # apps/api/tests/__init__.py
1284 # apps/api/tests/test_health.py
1285 # -*- coding: utf-8 -*-
1286 from __future__ import annotations
1287 import unittest
1288 class HealthTest(unittest.TestCase):
1289     def test_import_app(self) -> None:
1290         from apps.api.main import app
1291         self.assertIsNotNone(app)
1292     def test_health_endpoint(self) -> None:
1293         from fastapi.testclient import TestClient
1294         from apps.api.main import app
1295         c = TestClient(app)
1296         r = c.get("/api/health")
1297         self.assertEqual(r.status_code, 200)
1298         self.assertEqual(r.json().get("status"), "ok")
1299 if __name__ == "__main__":
1300     unittest.main()

```

```
1301 # apps/web/admin.html
1302 <!doctype html>
1303 <html lang="zh-CN"><head><meta charset="utf-8" /><meta name="viewport"
1304 content="width=device-width, initial-scale=1" />
1305 <title>管理员端 - 智能储能电池均衡控制系统</title></head><body style="font-
1306 family:Segoe UI,Microsoft YaHei,sans-serif; padding:18px;">
1307 <h1>管理员端</h1>
1308 <p>面向管理员：参数模板管理、角色权限、审计留痕与策略版本治理。</p>
1309 <ol>
1310 <li>维护阈值模板（电压差、温升、内阻漂移）。</li>
1311 <li>管理用户与角色授权，配置操作留痕策略。</li>
1312 <li>发布策略版本并维护回滚点。</li>
1313 </ol>
1314 </body></html>
1315 # apps/web/index.html
1316 <!doctype html>
1317 <html lang="zh-CN">
1318 <head>
1319 <meta charset="utf-8" />
1320 <meta name="viewport" content="width=device-width, initial-scale=1" />
1321 <title>智能储能电池均衡控制系统</title>
1322 <style>
1323 :root{--bg:#0b1020;--panel:#111a35;--text:#e9eeff;--muted:#a7b2da;--
1324 accent:#63f2d2;--warn:#ffb44a;--danger:#ff5b6e;}
1325 body{margin:0;font-family:Segoe UI,Microsoft YaHei,sans-serif;
1326 background:linear-gradient(180deg,#070a14,var(--bg));color:var(--text);}
1327 header{position:sticky;top:0;backdrop-filter:blur(10px);
1328 background:rgba(17,26,53,.78);border-bottom:1px solid rgba(99,242,210,.18);
1329 padding:14px 18px;display:flex;align-items:center;justify-content:space-
1330 between;gap:12px;}
1331 header .brand{font-weight:800;letter-spacing:.5px;}
1332 header .badge{font-size:12px;color:var(--accent);border:1px solid
1333 rgba(99,242,210,.35);padding:4px 10px;border-radius:999px;}
1334 main{max-width:1180px;margin:18px auto;padding:0 18px 28px;display:grid;
1335 grid-template-columns:260px 1fr;gap:16px;}
1336 nav{background:rgba(17,26,53,.72);border:1px solid rgba(99,242,210,.12);
1337 border-radius:14px;padding:12px;}
1338 nav a{display:block;color:var(--muted);text-decoration:none;padding:10px
1339 10px;border-radius:10px;}
1340 nav a:hover{background:rgba(99,242,210,.08);color:var(--text);}
1341 .panel{background:rgba(17,26,53,.72);border:1px solid rgba(99,242,210,
1342 .12);border-radius:14px;padding:14px;}
1343 .k{font-family:ui-monospace,SFMono-Regular,Menlo,Consolas,monospace;
1344 color:var(--accent);}
1345 .hidden{display:none !important;}
1346 .grid{display:grid;grid-template-columns:repeat(3,1fr);gap:12px;}
1347 .card{background:rgba(9,12,24,.55);border:1px solid rgba(99,242,210,.12);
1348 border-radius:12px;padding:12px;}
1349 .card h3{margin:0 0 8px;font-size:14px;color:var(--accent);}
1350 .hint{color:var(--muted);font-size:12px;line-height:1.6;}
```

```
1351     .row{display:grid;gap:10px;margin-top:10px;}
1352     input{padding:10px 12px;border-radius:10px;border:1px solid rgba(99,242,
1353 210,.18);background:rgba(255,255,255,.05);color:var(--text);}
1354     button{cursor:pointer;border-radius:10px;border:1px solid rgba(99,242,
1355 210,.22);background:rgba(99,242,210,.10);color:var(--text);padding:8px 12px;}
1356     button:hover{background:rgba(99,242,210,.16);}
1357     button.danger{border-color:rgba(255,91,110,.35);background:rgba(255,91,
1358 110,.12);}
1359     #modal{position:fixed;inset:0;background:rgba(0,0,0,.55);display:flex;
1360 align-items:center;justify-content:center;}
1361     #modal .dialog{width:min(720px,92vw);background:rgba(17,26,53,.96);
1362 border:1px solid rgba(99,242,210,.18);border-radius:14px;box-shadow:0 24px
1363 80px rgba(0,0,0,.45);padding:14px;}
1364     #modal header{position:static;border:none;background:transparent;
1365 padding:0 0 10px;}
1366     #modal .title{font-weight:800;}
1367 </style>
1368 </head>
1369 <body>
1370 <div id="loginView">
1371 <div style="max-width:520px;margin:10vh auto;padding:18px" class=
1372 "panel">
1373 <h1 style="margin:0 0 8px;font-size:18px;
1374 ">智能储能电池均衡控制系统</h1>
1375 <div class="hint">V1.0 演示环境：用于截图与材料生成的最小可运行界面。<
1376 /div>
1377 <form id="login-form" class="row">
1378 <label class="hint">用户名 <input name="username" value="admin"
1379 autocomplete="username" /></label>
1380 <label class="hint">密码 <input name="password" type="password"
1381 value="admin123" autocomplete="current-password" /></label>
1382 <button type="submit">登录</button>
1383 <div class="hint">提示：任意账号可登录（用于演示与截图）。</div>
1384 </form>
1385 </div>
1386 </div>
1387 <div id="appView" style="display:none;">
1388 <header>
1389 <div class="brand">电池均衡控制台</div>
1390 <div class="badge">V1.0</div>
1391 </header>
1392 <main>
1393 <nav>
1394 <a href="#home">首页</a>
1395 <a href="#packs">电池包</a>
1396 <a href="#cells">单体电芯</a>
1397 <a href="#balancing">均衡策略</a>
1398 <a href="#alarms">告警中心</a>
1399 <a href="#reports">统计报表</a>
1400 <a href="#audit">审计留痕</a>
```

```

1401     </nav>
1402     <section class="panel">
1403         <div style="display:flex;align-items:center;justify-content:space-
1404 between;gap:12px;">
1405             <h2 id="pageTitle" style="margin:0;font-size:16px;">首页</h2>
1406             <div class="hint">API: <span class="k" id=
1407 "apiStatus">未检测</span></div>
1408         </div>
1409         <div id="pageBody" style="margin-top:10px;"></div>
1410     </section>
1411 </main>
1412 </div>
1413 <div id="modal" class="modal-shell modal hidden">
1414     <div class="dialog" role="dialog" aria-modal="true">
1415         <header>
1416             <div class="title" id="modalTitle">操作</div>
1417             <button class="danger" id="modalClose">关闭</button>
1418         </header>
1419         <div class="hint" id="modalContent"></div>
1420     </div>
1421 </div>
1422 <script>
1423     const modal = document.getElementById('modal');
1424     const modalTitle = document.getElementById('modalTitle');
1425     const modalContent = document.getElementById('modalContent');
1426     function closeModal(){ modal.classList.add('hidden'); }
1427     function openModal(title, content){ modalTitle.textContent=title;
1428 modalContent.textContent=content; modal.classList.remove('hidden'); }
1429     document.getElementById('modalClose').onclick = closeModal;
1430     window.__rkModal = { closeModal, openModal };
1431     async function apiGet(p){ const r=await fetch(p); if(!r.ok) throw new
1432 Error('HTTP '+r.status); return await r.json(); }
1433     function setStatus(ok){ document.getElementById('apiStatus').textContent
1434 = ok ? 'OK' : 'FAIL'; }
1435     async function ping(){ try{ await apiGet('/api/health'); setStatus(true);
1436 } catch(e){ setStatus(false);} }
1437     function renderHome(){
1438         document.getElementById('pageTitle').textContent='首页';
1439         document.getElementById('pageBody').innerHTML = `
1440         <div class='grid'>
1441             <div class='card'><h3>均衡目标</h3><div class=
1442 'hint'>以“最小温升、最小内阻偏差、最短均衡时间”为约束，提升电芯一致性。</div>
1443 ></div>
1444             <div class='card'><h3>在线监测</h3><div class=
1445 'hint'>电压/温度/SoC/内阻偏差实时采集，并形成追溯报表。</div></div>
1446             <div class='card'><h3>闭环留痕</h3><div class=
1447 'hint'>策略下发、执行窗口、告警处置与复核结论自动记录。</div></div>
1448         </div>`;
1449     }
1450     function renderSimple(title, id){

```

```

1451     document.getElementById('pageTitle').textContent=title;
1452     const body = document.getElementById('pageBody');
1453     const descMap = {
1454         zones: '按站点/柜/簇划分资产边界，提供分区统计、权限隔离与设备定位。
1455     ',
1456         inspections: '对关键页面与接口执行健康检查，形成可追溯的校验记录。',
1457         alerts: '对电压差、温升、采样缺失等风险给出提醒，并引导处置闭环。',
1458         irrigation: '记录策略下发、执行窗口与回滚点，支撑版本治理与审计。',
1459         reports: '输出均衡效果与告警处置报表，支持导出与验收留痕。'
1460     };
1461     const desc = descMap[id] || '该模块用于展示业务信息与形成材料截图。';
1462     body.innerHTML = `<div id='${id}' class='card'><h3>${title}</h3><div
1463 class='hint'>${desc}</div><div class='hint'>模块标识: <span class='k'>#${id}
1464 </span></div></div>`;
1465     }
1466     function renderStations(){
1467         document.getElementById('pageTitle').textContent =
1468 '工作台（Stations）';
1469         const body = document.getElementById('pageBody');
1470         body.innerHTML = `
1471         <div id="stations" class="card">
1472             <h3>Stations 列表</h3>
1473             <div class="hint">用于演示 “新增/编辑/查看/删除” 弹窗操作按钮文案。<
1474 /div>
1475             <div class="toolbar">
1476                 <button onclick="window.__rkModal.openModal('新增',
1477 '创建新的均衡任务并填写字段。')">新增</button>
1478                 <button onclick="window.__rkModal.openModal('编辑',
1479 '修改任务阈值与执行窗口。')">编辑</button>
1480                 <button onclick="window.__rkModal.openModal('查看',
1481 '查看任务详情与引用关系。')">查看</button>
1482                 <button class="danger" onclick="window.__rkModal.openModal('删除
1483 ', '删除条目前请确认无引用并记录审计。')">删除</button>
1484             </div>
1485             <div class="hint">提示：请按业务流程依次执行新增、编辑、查看、删除
1486 ，并复核弹窗信息完整性。</div>
1487         </div>`;
1488     }
1489     function route(){
1490         const hash=(location.hash||'#home').replace(/^#/,"");
1491         if(hash==='home'||hash==='') return renderHome();
1492         if(hash==='zones') return renderSimple('资源分区','zones');
1493         if(hash==='stations') return renderStations();
1494         if(hash==='inspections') return renderSimple('交互检查','inspections');
1495         if(hash==='alerts') return renderSimple('风险提醒','alerts');
1496         if(hash==='irrigation') return renderSimple('发布流水','irrigation');
1497         if(hash==='reports') return renderSimple('统计报表','reports');
1498         return renderSimple('页面', hash);
1499     }
1500     document.getElementById('login-form').addEventListener('submit', (e)=>{

```

```
1501         e.preventDefault();
1502         localStorage.setItem('__demo_logged_in','1');
1503         document.getElementById('loginView').style.display='none';
1504         document.getElementById('appView').style.display='block';
1505         ping(); route();
1506     });
1507     window.addEventListener('hashchange', ()=>route());
1508     if(localStorage.getItem('__demo_logged_in')==='1'){
1509         document.getElementById('loginView').style.display='none';
1510         document.getElementById('appView').style.display='block';
1511         ping(); route();
1512     } else { renderHome(); }
1513 </script>
1514 </body>
1515 </html>
1516 # apps/web/student.html
1517 <!doctype html>
1518 <html lang="zh-CN"><head><meta charset="utf-8" /><meta name="viewport"
1519 content="width=device-width, initial-scale=1" />
1520 <title>值班员端 - 智能储能电池均衡控制系统</title></head><body style="font-
1521 family:Segoe UI,Microsoft YaHei,sans-serif; padding:18px;">
1522 <h1>值班员端</h1>
1523 <p>面向值班员：告警确认、策略下发、执行窗口控制与风险提示。</p>
1524 <ol>
1525     <li>确认告警并选择处置路径（观察/降载/停用）。</li>
1526     <li>查看均衡策略建议与预计温升。</li>
1527     <li>一键生成处置报告并提交复核。</li>
1528 </ol>
1529 </body></html>
1530 # apps/web/teacher.html
1531 <!doctype html>
1532 <html lang="zh-CN"><head><meta charset="utf-8" /><meta name="viewport"
1533 content="width=device-width, initial-scale=1" />
1534 <title>运行维护端 - 智能储能电池均衡控制系统</title></head><body style="font-
1535 family:Segoe UI,Microsoft YaHei,sans-serif; padding:18px;">
1536 <h1>运行维护端</h1>
1537 <p>面向站端运维：设备巡检、数据采样、阈值校准与工单闭环。</p>
1538 <ol>
1539     <li>查看采样窗口、采样频率与数据质量。</li>
1540     <li>对异常电芯做复核采样，并生成校准记录。</li>
1541     <li>导出运维日报与告警处置清单。</li>
1542 </ol>
1543 </body></html>
1544 # apps/api/main.py
1545 # -*- coding: utf-8 -*-
1546 from __future__ import annotations
1547 import os
1548 from pathlib import Path
1549 try:
1550     from fastapi import FastAPI
```

```

1551     from fastapi.responses import HTMLResponse
1552     from fastapi.staticfiles import StaticFiles
1553 except Exception: # pragma: no cover
1554     FastAPI = None # type: ignore[assignment]
1555 from apps.api.routers import health_router, packs_router
1556 APP_NAME = '智能储能电池均衡控制系统'
1557 def create_app() -> "FastAPI":
1558     if FastAPI is None:
1559         raise RuntimeError("FastAPI is not installed. Install dependencies
1560 before running the API.")
1561     app = FastAPI(title=APP_NAME)
1562     app.include_router(health_router)
1563     app.include_router(packs_router)
1564     web_root = Path(__file__).resolve().parents[1] / "web"
1565     app.mount("/static", StaticFiles(directory=str(web_root)), name="static")
1566     @app.get("/", response_class=HTMLResponse)
1567     def index() -> str:
1568         return (web_root / "index.html").read_text(encoding="utf-8")
1569     return app
1570 app = create_app()
1571 def _main() -> None:
1572     import uvicorn
1573     port = int(os.environ.get("APP_PORT", "8000"))
1574     uvicorn.run("apps.api.main:app", host="127.0.0.1", port=port, log_level=
1575 "warning")
1576 if __name__ == "__main__":
1577     _main()
1578 # apps/api/routers/_init_.py
1579 # -*- coding: utf-8 -*-
1580 from __future__ import annotations
1581 from .health import router as health_router
1582 from .packs import router as packs_router
1583 __all__ = ["health_router", "packs_router"]
1584 # apps/api/routers/health.py
1585 # -*- coding: utf-8 -*-
1586 from __future__ import annotations
1587 from fastapi import APIRouter
1588 router = APIRouter(prefix="/api", tags=["health"])
1589 @router.get("/health")
1590 def health() -> dict[str, str]:
1591     return {"status": "ok"}
1592 # apps/api/routers/packs.py
1593 # -*- coding: utf-8 -*-
1594 from __future__ import annotations
1595 import uuid
1596 from fastapi import APIRouter, Depends
1597 from apps.api.schemas.pack import PackCreate, PackView
1598 from apps.api.services.store import Store
1599 router = APIRouter(prefix="/api", tags=["packs"])
1600 def get_store() -> Store:

```



```

1601     project_root = __import__("pathlib").Path(__file__).resolve().parents[3]
1602     data_dir = project_root / "data"
1603     db_path = data_dir / "app.db"
1604     return Store(db_path=db_path)
1605 @router.get("/packs", response_model=list[PackView])
1606 def list_packs(store: Store = Depends(get_store)) -> list[PackView]:
1607     store.seed_demo()
1608     rows = store.list_packs()
1609     return [
1610         PackView(
1611             pack_id=r.pack_id,
1612             pack_code=r.pack_code,
1613             vendor=r.vendor,
1614             series_cells=r.series_cells,
1615             parallel_cells=r.parallel_cells,
1616             rated_capacity_ah=r.rated_capacity_ah,
1617             created_at=r.created_at,
1618         )
1619         for r in rows
1620     ]
1621 @router.post("/packs", response_model=dict[str, str])
1622 def create_pack(dto: PackCreate, store: Store = Depends(get_store)) ->
1623 dict[str, str]:
1624     pack_id = f"PACK-{uuid.uuid4().hex[:8].upper()}"
1625     store.create_pack(
1626         pack_id=pack_id,
1627         pack_code=dto.pack_code,
1628         vendor=dto.vendor,
1629         series_cells=dto.series_cells,
1630         parallel_cells=dto.parallel_cells,
1631         rated_capacity_ah=dto.rated_capacity_ah,
1632     )
1633     return {"pack_id": pack_id}
1634 # apps/api/domain/__init__.py
1635 # apps/api/domain/enums.py
1636 # -*- coding: utf-8 -*-
1637 from __future__ import annotations
1638 from enum import Enum
1639 class BalanceMode(str, Enum):
1640     passive = "passive"
1641     active = "active"
1642 class AlarmLevel(str, Enum):
1643     info = "info"
1644     warn = "warn"
1645     critical = "critical"
1646 # apps/api/services/__init__.py
1647 # apps/api/services/store.py
1648 # -*- coding: utf-8 -*-
1649 from __future__ import annotations
1650 import sqlite3

```

```
1651 from dataclasses import dataclass
1652 from datetime import datetime
1653 from pathlib import Path
1654 def utc_now() -> str:
1655     return datetime.utcnow().strftime("%Y-%m-%d %H:%M:%S")
1656 @dataclass(frozen=True)
1657 class PackRow:
1658     pack_id: str
1659     pack_code: str
1660     vendor: str
1661     series_cells: int
1662     parallel_cells: int
1663     rated_capacity_ah: float
1664     created_at: str
1665 class Store:
1666     def __init__(self, db_path: Path) -> None:
1667         self.db_path = db_path
1668         self.db_path.parent.mkdir(parents=True, exist_ok=True)
1669         self._init_db()
1670     def _connect(self) -> sqlite3.Connection:
1671         con = sqlite3.connect(str(self.db_path))
1672         con.row_factory = sqlite3.Row
1673         return con
1674     def _init_db(self) -> None:
1675         with self._connect() as con:
1676             con.execute(
1677                 """
1678                 create table if not exists packs(
1679                     pack_id text primary key,
1680                     pack_code text not null,
1681                     vendor text not null,
1682                     series_cells integer not null,
1683                     parallel_cells integer not null,
1684                     rated_capacity_ah real not null,
1685                     created_at text not null
1686                 );
1687                 """
1688             )
1689             con.commit()
1690     def seed_demo(self) -> None:
1691         rows = self.list_packs()
1692         if rows:
1693             return
1694         self.create_pack(
1695             pack_id="PACK-0001",
1696             pack_code="SZ-ESS-0001",
1697             vendor="示例能源",
1698             series_cells=96,
1699             parallel_cells=1,
1700             rated_capacity_ah=280.0,
```

```

1701     )
1702     self.create_pack(
1703         pack_id="PACK-0002",
1704         pack_code="SZ-ESS-0002",
1705         vendor="示例能源",
1706         series_cells=96,
1707         parallel_cells=1,
1708         rated_capacity_ah=280.0,
1709     )
1710     def create_pack(
1711         self,
1712         *,
1713         pack_id: str,
1714         pack_code: str,
1715         vendor: str,
1716         series_cells: int,
1717         parallel_cells: int,
1718         rated_capacity_ah: float,
1719     ) -> None:
1720         with self._connect() as con:
1721             con.execute(
1722                 """
1723                 insert into packs(pack_id, pack_code, vendor, series_cells,
1724 parallel_cells, rated_capacity_ah, created_at)
1725                 values(?,?,?,?,?,?,?)
1726                 """,
1727                 (pack_id, pack_code, vendor, int(series_cells),
1728 int(parallel_cells), float(rated_capacity_ah), utc_now()),
1729             )
1730             con.commit()
1731     def list_packs(self) -> list[PackRow]:
1732         with self._connect() as con:
1733             cur = con.execute(
1734                 """
1735                 select pack_id, pack_code, vendor, series_cells,
1736 parallel_cells, rated_capacity_ah, created_at
1737                 from packs order by created_at desc
1738                 """,
1739             )
1740             return [
1741                 PackRow(
1742                     pack_id=r["pack_id"],
1743                     pack_code=r["pack_code"],
1744                     vendor=r["vendor"],
1745                     series_cells=int(r["series_cells"]),
1746                     parallel_cells=int(r["parallel_cells"]),
1747                     rated_capacity_ah=float(r["rated_capacity_ah"]),
1748                     created_at=str(r["created_at"]),
1749                 )
1750                 for r in cur.fetchall()

```

```

1751     ]
1752 # apps/__init__.py
1753 # apps/api/__init__.py
1754 # apps/api/schemas/__init__.py
1755 # apps/api/schemas/pack.py
1756 # -*- coding: utf-8 -*-
1757 from __future__ import annotations
1758 from pydantic import BaseModel, Field
1759 class PackCreate(BaseModel):
1760     pack_code: str = Field(..., description="电池包编号")
1761     vendor: str = Field(..., description="厂家")
1762     series_cells: int = Field(..., ge=1, le=400, description="串联数量")
1763     parallel_cells: int = Field(..., ge=1, le=40, description="并联数量")
1764     rated_capacity_ah: float = Field(..., gt=0, description="额定容量(Ah)")
1765 class PackView(BaseModel):
1766     pack_id: str
1767     pack_code: str
1768     vendor: str
1769     series_cells: int
1770     parallel_cells: int
1771     rated_capacity_ah: float
1772     created_at: str
1773 # apps/api/tests/__init__.py
1774 # apps/api/tests/test_health.py
1775 # -*- coding: utf-8 -*-
1776 from __future__ import annotations
1777 import unittest
1778 class HealthTest(unittest.TestCase):
1779     def test_import_app(self) -> None:
1780         from apps.api.main import app
1781         self.assertIsNotNone(app)
1782     def test_health_endpoint(self) -> None:
1783         from fastapi.testclient import TestClient
1784         from apps.api.main import app
1785         c = TestClient(app)
1786         r = c.get("/api/health")
1787         self.assertEqual(r.status_code, 200)
1788         self.assertEqual(r.json().get("status"), "ok")
1789 if __name__ == "__main__":
1790     unittest.main()
1791 # apps/web/admin.html
1792 <!doctype html>
1793 <html lang="zh-CN"><head><meta charset="utf-8" /><meta name="viewport"
1794 content="width=device-width, initial-scale=1" />
1795 <title>管理员端 - 智能储能电池均衡控制系统</title></head><body style="font-
1796 family:Segoe UI,Microsoft YaHei,sans-serif; padding:18px;">
1797 <h1>管理员端</h1>
1798 <p>面向管理员：参数模板管理、角色权限、审计留痕与策略版本治理。</p>
1799 <ol>
1800 <li>维护阈值模板（电压差、温升、内阻漂移）。</li>

```

```
1801     <li>管理用户与角色授权，配置操作留痕策略。</li>
1802     <li>发布策略版本并维护回滚点。</li>
1803 </ol>
1804 </body></html>
1805 # apps/web/index.html
1806 <!doctype html>
1807 <html lang="zh-CN">
1808 <head>
1809     <meta charset="utf-8" />
1810     <meta name="viewport" content="width=device-width, initial-scale=1" />
1811     <title>智能储能电池均衡控制系统</title>
1812     <style>
1813         :root{--bg:#0b1020;--panel:#111a35;--text:#e9eeff;--muted:#a7b2da;--
1814         accent:#63f2d2;--warn:#ffb44a;--danger:#ff5b6e;}
1815         body{margin:0;font-family:Segoe UI,Microsoft YaHei,sans-serif;
1816         background:linear-gradient(180deg,#070a14,var(--bg));color:var(--text);}
1817         header{position:sticky;top:0;backdrop-filter:blur(10px);
1818         background:rgba(17,26,53,.78);border-bottom:1px solid rgba(99,242,210,.18);
1819         padding:14px 18px;display:flex;align-items:center;justify-content:space-
1820         between;gap:12px;}
1821         header .brand{font-weight:800;letter-spacing:.5px;}
1822         header .badge{font-size:12px;color:var(--accent);border:1px solid
1823         rgba(99,242,210,.35);padding:4px 10px;border-radius:999px;}
1824         main{max-width:1180px;margin:18px auto;padding:0 18px 28px;display:grid;
1825         grid-template-columns:260px 1fr;gap:16px;}
1826         nav{background:rgba(17,26,53,.72);border:1px solid rgba(99,242,210,.12);
1827         border-radius:14px;padding:12px;}
1828         nav a{display:block;color:var(--muted);text-decoration:none;padding:10px
1829         10px;border-radius:10px;}
1830         nav a:hover{background:rgba(99,242,210,.08);color:var(--text);}
1831         .panel{background:rgba(17,26,53,.72);border:1px solid rgba(99,242,210,
1832         .12);border-radius:14px;padding:14px;}
1833         .k{font-family:ui-monospace,SFMono-Regular,Menlo,Consolas,monospace;
1834         color:var(--accent);}
1835         .hidden{display:none !important;}
1836         .grid{display:grid;grid-template-columns:repeat(3,1fr);gap:12px;}
1837         .card{background:rgba(9,12,24,.55);border:1px solid rgba(99,242,210,.12);
1838         border-radius:12px;padding:12px;}
1839         .card h3{margin:0 0 8px;font-size:14px;color:var(--accent);}
1840         .hint{color:var(--muted);font-size:12px;line-height:1.6;}
1841         .row{display:grid;gap:10px;margin-top:10px;}
1842         input{padding:10px 12px;border-radius:10px;border:1px solid rgba(99,242,
1843         210,.18);background:rgba(255,255,255,.05);color:var(--text);}
1844         button{cursor:pointer;border-radius:10px;border:1px solid rgba(99,242,
1845         210,.22);background:rgba(99,242,210,.10);color:var(--text);padding:8px 12px;}
1846         button:hover{background:rgba(99,242,210,.16);}
1847         button.danger{border-color:rgba(255,91,110,.35);background:rgba(255,91,
1848         110,.12);}
1849         #modal{position:fixed;inset:0;background:rgba(0,0,0,.55);display:flex;
1850         align-items:center;justify-content:center;}
```

```
1851     #modal .dialog{width:min(720px,92vw);background:rgba(17,26,53,.96);
1852 border:1px solid rgba(99,242,210,.18);border-radius:14px;box-shadow:0 24px
1853 80px rgba(0,0,0,.45);padding:14px;}
1854     #modal header{position:static;border:none;background:transparent;
1855 padding:0 0 10px;}
1856     #modal .title{font-weight:800;}
1857     </style>
1858 </head>
1859 <body>
1860     <div id="loginView">
1861     <div style="max-width:520px;margin:10vh auto;padding:18px" class=
1862 "panel">
1863         <h1 style="margin:0 0 8px;font-size:18px;
1864 ">智能储能电池均衡控制系统</h1>
1865         <div class="hint">V1.0 演示环境：用于截图与材料生成的最小可运行界面。<
1866 /div>
1867         <form id="login-form" class="row">
1868             <label class="hint">用户名 <input name="username" value="admin"
1869 autocomplete="username" /></label>
1870             <label class="hint">密码 <input name="password" type="password"
1871 value="admin123" autocomplete="current-password" /></label>
1872             <button type="submit">登录</button>
1873             <div class="hint">提示：任意账号可登录（用于演示与截图）。</div>
1874         </form>
1875     </div>
1876 </div>
1877 <div id="appView" style="display:none;">
1878     <header>
1879         <div class="brand">电池均衡控制台</div>
1880         <div class="badge">V1.0</div>
1881     </header>
1882     <main>
1883     <nav>
1884         <a href="#home">首页</a>
1885         <a href="#packs">电池包</a>
1886         <a href="#cells">单体电芯</a>
1887         <a href="#balancing">均衡策略</a>
1888         <a href="#alarms">告警中心</a>
1889         <a href="#reports">统计报表</a>
1890         <a href="#audit">审计留痕</a>
1891     </nav>
1892     <section class="panel">
1893         <div style="display:flex;align-items:center;justify-content:space-
1894 between;gap:12px;">
1895             <h2 id="pageTitle" style="margin:0;font-size:16px;">首页</h2>
1896             <div class="hint">API: <span class="k" id=
1897 "apiStatus">未检测</span></div>
1898         </div>
1899         <div id="pageBody" style="margin-top:10px;"></div>
1900     </section>
```

```

1901     </main>
1902 </div>
1903 <div id="modal" class="modal-shell modal hidden">
1904     <div class="dialog" role="dialog" aria-modal="true">
1905         <header>
1906             <div class="title" id="modalTitle">操作</div>
1907             <button class="danger" id="modalClose">关闭</button>
1908         </header>
1909         <div class="hint" id="modalContent"></div>
1910     </div>
1911 </div>
1912 <script>
1913     const modal = document.getElementById('modal');
1914     const modalTitle = document.getElementById('modalTitle');
1915     const modalContent = document.getElementById('modalContent');
1916     function closeModal(){ modal.classList.add('hidden'); }
1917     function openModal(title, content){ modalTitle.textContent=title;
1918 modalContent.textContent=content; modal.classList.remove('hidden'); }
1919     document.getElementById('modalClose').onclick = closeModal;
1920     window.__rkModal = { closeModal, openModal };
1921     async function apiGet(p){ const r=await fetch(p); if(!r.ok) throw new
1922 Error('HTTP '+r.status); return await r.json(); }
1923     function setStatus(ok){ document.getElementById('apiStatus').textContent
1924 = ok ? 'OK' : 'FAIL'; }
1925     async function ping(){ try{ await apiGet('/api/health'); setStatus(true);
1926 } catch(e){ setStatus(false);} }
1927     function renderHome(){
1928         document.getElementById('pageTitle').textContent='首页';
1929         document.getElementById('pageBody').innerHTML = `
1930         <div class='grid'>
1931             <div class='card'><h3>均衡目标</h3><div class=
1932 'hint'>以“最小温升、最小内阻偏差、最短均衡时间”为约束，提升电芯一致性。</div>
1933 ></div>
1934             <div class='card'><h3>在线监测</h3><div class=
1935 'hint'>电压/温度/Soc/内阻偏差实时采集，并形成追溯报表。</div></div>
1936             <div class='card'><h3>闭环留痕</h3><div class=
1937 'hint'>策略下发、执行窗口、告警处置与复核结论自动记录。</div></div>
1938         </div>`;
1939     }
1940     function renderSimple(title, id){
1941         document.getElementById('pageTitle').textContent=title;
1942         const body = document.getElementById('pageBody');
1943         const descMap = {
1944             zones: '按站点/柜/簇划分资产边界，提供分区统计、权限隔离与设备定位。
1945 ',
1946             inspections: '对关键页面与接口执行健康检查，形成可追溯的校验记录。',
1947             alerts: '对电压差、温升、采样缺失等风险给出提醒，并引导处置闭环。',
1948             irrigation: '记录策略下发、执行窗口与回滚点，支撑版本治理与审计。',
1949             reports: '输出均衡效果与告警处置报表，支持导出与验收留痕。'
1950         };

```

```
1951     const desc = descMap[id] || '该模块用于展示业务信息与形成材料截图。';
1952     body.innerHTML = `<div id='${id}' class='card'><h3>${title}</h3><div
1953 class='hint'>${desc}</div><div class='hint'>模块标识: <span class='k'>#${id}
1954 </span></div></div>`;
1955     }
1956     function renderStations(){
1957         document.getElementById('pageTitle').textContent =
1958 '工作台（Stations）';
1959         const body = document.getElementById('pageBody');
1960         body.innerHTML = `
1961         <div id="stations" class="card">
1962             <h3>Stations 列表</h3>
1963             <div class="hint">用于演示 “新增/编辑/查看/删除” 弹窗操作按钮文案。<
1964 /div>
1965             <div class="toolbar">
1966                 <button onclick="window.__rkModal.openModal('新增',
1967 '创建新的均衡任务并填写字段。')">新增</button>
1968                 <button onclick="window.__rkModal.openModal('编辑',
1969 '修改任务阈值与执行窗口。')">编辑</button>
1970                 <button onclick="window.__rkModal.openModal('查看',
1971 '查看任务详情与引用关系。')">查看</button>
1972                 <button class="danger" onclick="window.__rkModal.openModal('删除
1973 ', '删除条目前请确认无引用并记录审计。')">删除</button>
1974             </div>
1975             <div class="hint">提示：请按业务流程依次执行新增、编辑、查看、删除
1976 ，并复核弹窗信息完整性。</div>
1977             </div>`;
1978     }
1979     function route(){
1980         const hash=(location.hash||"#home").replace(/^#/,"");
1981         if(hash=="home"||hash=="") return renderHome();
1982         if(hash=="zones") return renderSimple('资源分区','zones');
1983         if(hash=="stations") return renderStations();
1984         if(hash=="inspections") return renderSimple('交互检查','inspections');
1985         if(hash=="alerts") return renderSimple('风险提示','alerts');
1986         if(hash=="irrigation") return renderSimple('发布流水','irrigation');
1987         if(hash=="reports") return renderSimple('统计报表','reports');
1988         return renderSimple('页面', hash);
1989     }
1990     document.getElementById('login-form').addEventListener('submit', (e)=>{
1991         e.preventDefault();
1992         localStorage.setItem('__demo_logged_in','1');
1993         document.getElementById('loginView').style.display='none';
1994         document.getElementById('appView').style.display='block';
1995         ping(); route();
1996     });
1997     window.addEventListener('hashchange', ()=>route());
1998     if(localStorage.getItem('__demo_logged_in')=="1"){
1999         document.getElementById('loginView').style.display='none';
2000         document.getElementById('appView').style.display='block';
```



```
2001         ping(); route();
2002     } else { renderHome(); }
2003 </script>
2004 </body>
2005 </html>
2006 # apps/web/student.html
2007 <!doctype html>
2008 <html lang="zh-CN"><head><meta charset="utf-8" /><meta name="viewport"
2009 content="width=device-width, initial-scale=1" />
2010 <title>值班员端 - 智能储能电池均衡控制系统</title></head><body style="font-
2011 family:Segoe UI,Microsoft YaHei,sans-serif; padding:18px;">
2012 <h1>值班员端</h1>
2013 <p>面向值班员：告警确认、策略下发、执行窗口控制与风险提示。</p>
2014 <ol>
2015     <li>确认告警并选择处置路径（观察/降载/停用）。</li>
2016     <li>查看均衡策略建议与预计温升。</li>
2017     <li>一键生成处置报告并提交复核。</li>
2018 </ol>
2019 </body></html>
2020 # apps/web/teacher.html
2021 <!doctype html>
2022 <html lang="zh-CN"><head><meta charset="utf-8" /><meta name="viewport"
2023 content="width=device-width, initial-scale=1" />
2024 <title>运行维护端 - 智能储能电池均衡控制系统</title></head><body style="font-
2025 family:Segoe UI,Microsoft YaHei,sans-serif; padding:18px;">
2026 <h1>运行维护端</h1>
2027 <p>面向站端运维：设备巡检、数据采样、阈值校准与工单闭环。</p>
2028 <ol>
2029     <li>查看采样窗口、采样频率与数据质量。</li>
2030     <li>对异常电芯做复核采样，并生成校准记录。</li>
2031     <li>导出运维日报与告警处置清单。</li>
2032 </ol>
2033 </body></html>
2034 # apps/api/main.py
2035 # -*- coding: utf-8 -*-
2036 from __future__ import annotations
2037 import os
2038 from pathlib import Path
2039 try:
2040     from fastapi import FastAPI
2041     from fastapi.responses import HTMLResponse
2042     from fastapi.staticfiles import StaticFiles
2043 except Exception: # pragma: no cover
2044     FastAPI = None # type: ignore[assignment]
2045 from apps.api.routers import health_router, packs_router
2046 APP_NAME = '智能储能电池均衡控制系统'
2047 def create_app() -> "FastAPI":
2048     if FastAPI is None:
2049         raise RuntimeError("FastAPI is not installed. Install dependencies
2050 before running the API.")
```

```

2051     app = FastAPI(title=APP_NAME)
2052     app.include_router(health_router)
2053     app.include_router(packs_router)
2054     web_root = Path(__file__).resolve().parents[1] / "web"
2055     app.mount("/static", StaticFiles(directory=str(web_root)), name="static")
2056     @app.get("/", response_class=HTMLResponse)
2057     def index() -> str:
2058         return (web_root / "index.html").read_text(encoding="utf-8")
2059     return app
2060 app = create_app()
2061 def _main() -> None:
2062     import uvicorn
2063     port = int(os.environ.get("APP_PORT", "8000"))
2064     uvicorn.run("apps.api.main:app", host="127.0.0.1", port=port, log_level=
2065 "warning")
2066 if __name__ == "__main__":
2067     _main()
2068 # apps/api/routers/_init_.py
2069 # -*- coding: utf-8 -*-
2070 from __future__ import annotations
2071 from .health import router as health_router
2072 from .packs import router as packs_router
2073 __all__ = ["health_router", "packs_router"]
2074 # apps/api/routers/health.py
2075 # -*- coding: utf-8 -*-
2076 from __future__ import annotations
2077 from fastapi import APIRouter
2078 router = APIRouter(prefix="/api", tags=["health"])
2079 @router.get("/health")
2080 def health() -> dict[str, str]:
2081     return {"status": "ok"}
2082 # apps/api/routers/packs.py
2083 # -*- coding: utf-8 -*-
2084 from __future__ import annotations
2085 import uuid
2086 from fastapi import APIRouter, Depends
2087 from apps.api.schemas.pack import PackCreate, PackView
2088 from apps.api.services.store import Store
2089 router = APIRouter(prefix="/api", tags=["packs"])
2090 def get_store() -> Store:
2091     project_root = __import__("pathlib").Path(__file__).resolve().parents[3]
2092     data_dir = project_root / "data"
2093     db_path = data_dir / "app.db"
2094     return Store(db_path=db_path)
2095 @router.get("/packs", response_model=list[PackView])
2096 def list_packs(store: Store = Depends(get_store)) -> list[PackView]:
2097     store.seed_demo()
2098     rows = store.list_packs()
2099     return [
2100         PackView(

```

```

2101         pack_id=r.pack_id,
2102         pack_code=r.pack_code,
2103         vendor=r.vendor,
2104         series_cells=r.series_cells,
2105         parallel_cells=r.parallel_cells,
2106         rated_capacity_ah=r.rated_capacity_ah,
2107         created_at=r.created_at,
2108     )
2109     for r in rows
2110 ]
2111 @router.post("/packs", response_model=dict[str, str])
2112 def create_pack(dto: PackCreate, store: Store = Depends(get_store)) ->
2113 dict[str, str]:
2114     pack_id = f"PACK-{uuid.uuid4().hex[:8].upper()}"
2115     store.create_pack(
2116         pack_id=pack_id,
2117         pack_code=dto.pack_code,
2118         vendor=dto.vendor,
2119         series_cells=dto.series_cells,
2120         parallel_cells=dto.parallel_cells,
2121         rated_capacity_ah=dto.rated_capacity_ah,
2122     )
2123     return {"pack_id": pack_id}
2124 # apps/api/domain/__init__.py
2125 # apps/api/domain/enums.py
2126 # -*- coding: utf-8 -*-
2127 from __future__ import annotations
2128 from enum import Enum
2129 class BalanceMode(str, Enum):
2130     passive = "passive"
2131     active = "active"
2132 class AlarmLevel(str, Enum):
2133     info = "info"
2134     warn = "warn"
2135     critical = "critical"
2136 # apps/api/services/__init__.py
2137 # apps/api/services/store.py
2138 # -*- coding: utf-8 -*-
2139 from __future__ import annotations
2140 import sqlite3
2141 from dataclasses import dataclass
2142 from datetime import datetime
2143 from pathlib import Path
2144 def utc_now() -> str:
2145     return datetime.utcnow().strftime("%Y-%m-%d %H:%M:%S")
2146 @dataclass(frozen=True)
2147 class PackRow:
2148     pack_id: str
2149     pack_code: str
2150     vendor: str

```

```
2151     series_cells: int
2152     parallel_cells: int
2153     rated_capacity_ah: float
2154     created_at: str
2155 class Store:
2156     def __init__(self, db_path: Path) -> None:
2157         self.db_path = db_path
2158         self.db_path.parent.mkdir(parents=True, exist_ok=True)
2159         self._init_db()
2160     def _connect(self) -> sqlite3.Connection:
2161         con = sqlite3.connect(str(self.db_path))
2162         con.row_factory = sqlite3.Row
2163         return con
2164     def _init_db(self) -> None:
2165         with self._connect() as con:
2166             con.execute(
2167                 """
2168                 create table if not exists packs(
2169                     pack_id text primary key,
2170                     pack_code text not null,
2171                     vendor text not null,
2172                     series_cells integer not null,
2173                     parallel_cells integer not null,
2174                     rated_capacity_ah real not null,
2175                     created_at text not null
2176                 );
2177                 """
2178             )
2179             con.commit()
2180     def seed_demo(self) -> None:
2181         rows = self.list_packs()
2182         if rows:
2183             return
2184         self.create_pack(
2185             pack_id="PACK-0001",
2186             pack_code="SZ-ESS-0001",
2187             vendor="示例能源",
2188             series_cells=96,
2189             parallel_cells=1,
2190             rated_capacity_ah=280.0,
2191         )
2192         self.create_pack(
2193             pack_id="PACK-0002",
2194             pack_code="SZ-ESS-0002",
2195             vendor="示例能源",
2196             series_cells=96,
2197             parallel_cells=1,
2198             rated_capacity_ah=280.0,
2199         )
2200     def create_pack(
```

```

2201     self,
2202     *,
2203     pack_id: str,
2204     pack_code: str,
2205     vendor: str,
2206     series_cells: int,
2207     parallel_cells: int,
2208     rated_capacity_ah: float,
2209 ) -> None:
2210     with self._connect() as con:
2211         con.execute(
2212             """
2213             insert into packs(pack_id, pack_code, vendor, series_cells,
2214 parallel_cells, rated_capacity_ah, created_at)
2215             values(?,?,?,?,?,?,?)
2216             """,
2217             (pack_id, pack_code, vendor, int(series_cells),
2218 int(parallel_cells), float(rated_capacity_ah), utc_now()),
2219             )
2220         con.commit()
2221     def list_packs(self) -> list[PackRow]:
2222         with self._connect() as con:
2223             cur = con.execute(
2224                 """
2225                 select pack_id, pack_code, vendor, series_cells,
2226 parallel_cells, rated_capacity_ah, created_at
2227                 from packs order by created_at desc
2228                 """
2229             )
2230             return [
2231                 PackRow(
2232                     pack_id=r["pack_id"],
2233                     pack_code=r["pack_code"],
2234                     vendor=r["vendor"],
2235                     series_cells=int(r["series_cells"]),
2236                     parallel_cells=int(r["parallel_cells"]),
2237                     rated_capacity_ah=float(r["rated_capacity_ah"]),
2238                     created_at=str(r["created_at"]),
2239                 )
2240                 for r in cur.fetchall()
2241             ]
2242 # apps/__init__.py
2243 # apps/api/__init__.py
2244 # apps/api/schemas/__init__.py
2245 # apps/api/schemas/pack.py
2246 # -*- coding: utf-8 -*-
2247 from __future__ import annotations
2248 from pydantic import BaseModel, Field
2249 class PackCreate(BaseModel):
2250     pack_code: str = Field(..., description="电池包编号")

```

```

2251     vendor: str = Field(..., description="厂家")
2252     series_cells: int = Field(..., ge=1, le=400, description="串联数量")
2253     parallel_cells: int = Field(..., ge=1, le=40, description="并联数量")
2254     rated_capacity_ah: float = Field(..., gt=0, description="额定容量(Ah)")
2255 class PackView(BaseModel):
2256     pack_id: str
2257     pack_code: str
2258     vendor: str
2259     series_cells: int
2260     parallel_cells: int
2261     rated_capacity_ah: float
2262     created_at: str
2263 # apps/api/tests/__init__.py
2264 # apps/api/tests/test_health.py
2265 # -*- coding: utf-8 -*-
2266 from __future__ import annotations
2267 import unittest
2268 class HealthTest(unittest.TestCase):
2269     def test_import_app(self) -> None:
2270         from apps.api.main import app
2271         self.assertIsNotNone(app)
2272     def test_health_endpoint(self) -> None:
2273         from fastapi.testclient import TestClient
2274         from apps.api.main import app
2275         c = TestClient(app)
2276         r = c.get("/api/health")
2277         self.assertEqual(r.status_code, 200)
2278         self.assertEqual(r.json().get("status"), "ok")
2279 if __name__ == "__main__":
2280     unittest.main()
2281 # apps/web/admin.html
2282 <!doctype html>
2283 <html lang="zh-CN"><head><meta charset="utf-8" /><meta name="viewport"
2284 content="width=device-width, initial-scale=1" />
2285 <title>管理员端 - 智能储能电池均衡控制系统</title></head><body style="font-
2286 family: Segoe UI, Microsoft YaHei, sans-serif; padding: 18px;">
2287 <h1>管理员端</h1>
2288 <p>面向管理员：参数模板管理、角色权限、审计留痕与策略版本治理。</p>
2289 <ol>
2290     <li>维护阈值模板（电压差、温升、内阻漂移）。</li>
2291     <li>管理用户与角色授权，配置操作留痕策略。</li>
2292     <li>发布策略版本并维护回滚点。</li>
2293 </ol>
2294 </body></html>
2295 # apps/web/index.html
2296 <!doctype html>
2297 <html lang="zh-CN">
2298 <head>
2299     <meta charset="utf-8" />
2300     <meta name="viewport" content="width=device-width, initial-scale=1" />

```

```
2301 <title>智能储能电池均衡控制系统</title>
2302 <style>
2303   :root{--bg:#0b1020;--panel:#111a35;--text:#e9eeff;--muted:#a7b2da;--
2304   accent:#63f2d2;--warn:#ffb44a;--danger:#ff5b6e;}
2305   body{margin:0;font-family:Segoe UI,Microsoft YaHei,sans-serif;
2306   background:linear-gradient(180deg,#070a14,var(--bg));color:var(--text);}
2307   header{position:sticky;top:0;backdrop-filter:blur(10px);
2308   background:rgba(17,26,53,.78);border-bottom:1px solid rgba(99,242,210,.18);
2309   padding:14px 18px;display:flex;align-items:center;justify-content:space-
2310   between;gap:12px;}
2311   header .brand{font-weight:800;letter-spacing:.5px;}
2312   header .badge{font-size:12px;color:var(--accent);border:1px solid
2313   rgba(99,242,210,.35);padding:4px 10px;border-radius:999px;}
2314   main{max-width:1180px;margin:18px auto;padding:0 18px 28px;display:grid;
2315   grid-template-columns:260px 1fr;gap:16px;}
2316   nav{background:rgba(17,26,53,.72);border:1px solid rgba(99,242,210,.12);
2317   border-radius:14px;padding:12px;}
2318   nav a{display:block;color:var(--muted);text-decoration:none;padding:10px
2319   10px;border-radius:10px;}
2320   nav a:hover{background:rgba(99,242,210,.08);color:var(--text);}
2321   .panel{background:rgba(17,26,53,.72);border:1px solid rgba(99,242,210,
2322   .12);border-radius:14px;padding:14px;}
2323   .k{font-family:ui-monospace,SFMono-Regular,Menlo,Consolas,monospace;
2324   color:var(--accent);}
2325   .hidden{display:none !important;}
2326   .grid{display:grid;grid-template-columns:repeat(3,1fr);gap:12px;}
2327   .card{background:rgba(9,12,24,.55);border:1px solid rgba(99,242,210,.12);
2328   border-radius:12px;padding:12px;}
2329   .card h3{margin:0 0 8px;font-size:14px;color:var(--accent);}
2330   .hint{color:var(--muted);font-size:12px;line-height:1.6;}
2331   .row{display:grid;gap:10px;margin-top:10px;}
2332   input{padding:10px 12px;border-radius:10px;border:1px solid rgba(99,242,
2333   210,.18);background:rgba(255,255,255,.05);color:var(--text);}
2334   button{cursor:pointer;border-radius:10px;border:1px solid rgba(99,242,
2335   210,.22);background:rgba(99,242,210,.10);color:var(--text);padding:8px 12px;}
2336   button:hover{background:rgba(99,242,210,.16);}
2337   button.danger{border-color:rgba(255,91,110,.35);background:rgba(255,91,
2338   110,.12);}
2339   #modal{position:fixed;inset:0;background:rgba(0,0,0,.55);display:flex;
2340   align-items:center;justify-content:center;}
2341   #modal .dialog{width:min(720px,92vw);background:rgba(17,26,53,.96);
2342   border:1px solid rgba(99,242,210,.18);border-radius:14px;box-shadow:0 24px
2343   80px rgba(0,0,0,.45);padding:14px;}
2344   #modal header{position:static;border:none;background:transparent;
2345   padding:0 0 10px;}
2346   #modal .title{font-weight:800;}
2347 </style>
2348 </head>
2349 <body>
2350 <div id="loginView">
```

```
2351     <div style="max-width:520px;margin:10vh auto;padding:18px" class=
2352 "panel">
2353     <h1 style="margin:0 0 8px;font-size:18px;
2354 ">智能储能电池均衡控制系统</h1>
2355     <div class="hint">V1.0 演示环境：用于截图与材料生成的最小可运行界面。<
2356 /div>
2357     <form id="login-form" class="row">
2358     <label class="hint">用户名 <input name="username" value="admin"
2359 autocomplete="username" /></label>
2360     <label class="hint">密码 <input name="password" type="password"
2361 value="admin123" autocomplete="current-password" /></label>
2362     <button type="submit">登录</button>
2363     <div class="hint">提示：任意账号可登录（用于演示与截图）。</div>
2364     </form>
2365 </div>
2366 </div>
2367 <div id="appView" style="display:none;">
2368 <header>
2369     <div class="brand">电池均衡控制台</div>
2370     <div class="badge">V1.0</div>
2371 </header>
2372 <main>
2373 <nav>
2374     <a href="#home">首页</a>
2375     <a href="#packs">电池包</a>
2376     <a href="#cells">单体电芯</a>
2377     <a href="#balancing">均衡策略</a>
2378     <a href="#alarms">告警中心</a>
2379     <a href="#reports">统计报表</a>
2380     <a href="#audit">审计留痕</a>
2381 </nav>
2382 <section class="panel">
2383     <div style="display:flex;align-items:center;justify-content:space-
2384 between;gap:12px;">
2385         <h2 id="pageTitle" style="margin:0;font-size:16px;">首页</h2>
2386         <div class="hint">API: <span class="k" id=
2387 "apiStatus">未检测</span></div>
2388     </div>
2389     <div id="pageBody" style="margin-top:10px;"></div>
2390 </section>
2391 </main>
2392 </div>
2393 <div id="modal" class="modal-shell modal hidden">
2394 <div class="dialog" role="dialog" aria-modal="true">
2395 <header>
2396     <div class="title" id="modalTitle">操作</div>
2397     <button class="danger" id="modalClose">关闭</button>
2398 </header>
2399 <div class="hint" id="modalContent"></div>
2400 </div>
```



```

2401     </div>
2402     <script>
2403         const modal = document.getElementById('modal');
2404         const modalTitle = document.getElementById('modalTitle');
2405         const modalContent = document.getElementById('modalContent');
2406         function closeModal(){ modal.classList.add('hidden'); }
2407         function openModal(title, content){ modalTitle.textContent=title;
2408 modalContent.textContent=content; modal.classList.remove('hidden'); }
2409         document.getElementById('modalClose').onclick = closeModal;
2410         window.__rkModal = { closeModal, openModal };
2411         async function apiGet(p){ const r=await fetch(p); if(!r.ok) throw new
2412 Error('HTTP '+r.status); return await r.json(); }
2413         function setStatus(ok){ document.getElementById('apiStatus').textContent
2414 = ok ? 'OK' : 'FAIL'; }
2415         async function ping(){ try{ await apiGet('/api/health'); setStatus(true);
2416 } catch(e){ setStatus(false);} }
2417         function renderHome(){
2418             document.getElementById('pageTitle').textContent='首页';
2419             document.getElementById('pageBody').innerHTML = `
2420                 <div class='grid'>
2421                     <div class='card'><h3>均衡目标</h3><div class=
2422 'hint'>以 “最小温升、最小内阻偏差、最短均衡时间” 为约束，提升电芯一致性。</div>
2423                 <</div>
2424                     <div class='card'><h3>在线监测</h3><div class=
2425 'hint'>电压/温度/Soc/内阻偏差实时采集，并形成追溯报表。</div></div>
2426                     <div class='card'><h3>闭环留痕</h3><div class=
2427 'hint'>策略下发、执行窗口、告警处置与复核结论自动记录。</div></div>
2428                 </div>`;
2429             }
2430             function renderSimple(title, id){
2431                 document.getElementById('pageTitle').textContent=title;
2432                 const body = document.getElementById('pageBody');
2433                 const descMap = {
2434                     zones: '按站点/柜/簇划分资产边界，提供分区统计、权限隔离与设备定位。',
2435                     inspections: '对关键页面与接口执行健康检查，形成可追溯的校验记录。',
2436                     alerts: '对电压差、温升、采样缺失等风险给出提醒，并引导处置闭环。',
2437                     irrigation: '记录策略下发、执行窗口与回滚点，支撑版本治理与审计。',
2438                     reports: '输出均衡效果与告警处置报表，支持导出与验收留痕。'
2439                 };
2440                 const desc = descMap[id] || '该模块用于展示业务信息与形成材料截图。';
2441                 body.innerHTML = `<div id='${id}' class='card'><h3>${title}</h3><div
2442 class='hint'>${desc}</div><div class='hint'>模块标识: <span class='k'>#${id}
2443 </span></div></div>`;
2444             }
2445             function renderStations(){
2446                 document.getElementById('pageTitle').textContent =
2447 '工作台（Stations）';
2448                 const body = document.getElementById('pageBody');
2449                 body.innerHTML = `
2450 
```

```

2451     <div id="stations" class="card">
2452         <h3>Stations 列表</h3>
2453         <div class="hint">用于演示 “新增/编辑/查看/删除” 弹窗操作按钮文案。<
2454 /div>
2455         <div class="toolbar">
2456             <button onclick="window.__rkModal.openModal('新增',
2457 '创建新的均衡任务并填写字段。')">新增</button>
2458             <button onclick="window.__rkModal.openModal('编辑',
2459 '修改任务阈值与执行窗口。')">编辑</button>
2460             <button onclick="window.__rkModal.openModal('查看',
2461 '查看任务详情与引用关系。')">查看</button>
2462             <button class="danger" onclick="window.__rkModal.openModal('删除
2463 ', '删除条目前请确认无引用并记录审计。')">删除</button>
2464         </div>
2465         <div class="hint">提示：请按业务流程依次执行新增、编辑、查看、删除
2466 ，并复核弹窗信息完整性。</div>
2467     </div>;
2468 }
2469 function route(){
2470     const hash=(location.hash||'#home').replace(/^#/, "");
2471     if(hash==='home'||hash==='') return renderHome();
2472     if(hash==='zones') return renderSimple('资源分区','zones');
2473     if(hash==='stations') return renderStations();
2474     if(hash==='inspections') return renderSimple('交互检查','inspections');
2475     if(hash==='alerts') return renderSimple('风险提醒','alerts');
2476     if(hash==='irrigation') return renderSimple('发布流水','irrigation');
2477     if(hash==='reports') return renderSimple('统计报表','reports');
2478     return renderSimple('页面', hash);
2479 }
2480 document.getElementById('login-form').addEventListener('submit', (e)=>{
2481     e.preventDefault();
2482     localStorage.setItem('__demo_logged_in','1');
2483     document.getElementById('loginView').style.display='none';
2484     document.getElementById('appView').style.display='block';
2485     ping(); route();
2486 });
2487 window.addEventListener('hashchange', ()=>route());
2488 if(localStorage.getItem('__demo_logged_in')==='1'){
2489     document.getElementById('loginView').style.display='none';
2490     document.getElementById('appView').style.display='block';
2491     ping(); route();
2492 } else { renderHome(); }
2493 </script>
2494 </body>
2495 </html>
2496 # apps/web/student.html
2497 <!doctype html>
2498 <html lang="zh-CN"><head><meta charset="utf-8" /><meta name="viewport"
2499 content="width=device-width, initial-scale=1" />
2500 <title>值班员端 - 智能储能电池均衡控制系统</title></head><body style="font-

```

```
2501     family:Segoe UI,Microsoft YaHei,sans-serif; padding:18px;">
2502 <h1>值班员端</h1>
2503 <p>面向值班员：告警确认、策略下发、执行窗口控制与风险提示。</p>
2504 <ol>
2505     <li>确认告警并选择处置路径（观察/降载/停用）。</li>
2506     <li>查看均衡策略建议与预计温升。</li>
2507     <li>一键生成处置报告并提交复核。</li>
2508 </ol>
2509 </body></html>
2510 # apps/web/teacher.html
2511 <!doctype html>
2512 <html lang="zh-CN"><head><meta charset="utf-8" /><meta name="viewport"
2513 content="width=device-width, initial-scale=1" />
2514 <title>运行维护端 - 智能储能电池均衡控制系统</title></head><body style="font-
2515 family:Segoe UI,Microsoft YaHei,sans-serif; padding:18px;">
2516 <h1>运行维护端</h1>
2517 <p>面向站端运维：设备巡检、数据采样、阈值校准与工单闭环。</p>
2518 <ol>
2519     <li>查看采样窗口、采样频率与数据质量。</li>
2520     <li>对异常电芯做复核采样，并生成校准记录。</li>
2521     <li>导出运维日报与告警处置清单。</li>
2522 </ol>
2523 </body></html>
2524 # apps/api/main.py
2525 # -*- coding: utf-8 -*-
2526 from __future__ import annotations
2527 import os
2528 from pathlib import Path
2529 try:
2530     from fastapi import FastAPI
2531     from fastapi.responses import HTMLResponse
2532     from fastapi.staticfiles import StaticFiles
2533 except Exception: # pragma: no cover
2534     FastAPI = None # type: ignore[assignment]
2535 from apps.api.routers import health_router, packs_router
2536 APP_NAME = '智能储能电池均衡控制系统'
2537 def create_app() -> "FastAPI":
2538     if FastAPI is None:
2539         raise RuntimeError("FastAPI is not installed. Install dependencies
2540 before running the API.")
2541     app = FastAPI(title=APP_NAME)
2542     app.include_router(health_router)
2543     app.include_router(packs_router)
2544     web_root = Path(__file__).resolve().parents[1] / "web"
2545     app.mount("/static", StaticFiles(directory=str(web_root)), name="static")
2546     @app.get("/", response_class=HTMLResponse)
2547     def index() -> str:
2548         return (web_root / "index.html").read_text(encoding="utf-8")
2549     return app
2550 app = create_app()
```

```

2551 def _main() -> None:
2552     import uvicorn
2553     port = int(os.environ.get("APP_PORT", "8000"))
2554     uvicorn.run("apps.api.main:app", host="127.0.0.1", port=port, log_level=
2555 "warning")
2556 if __name__ == "__main__":
2557     _main()
2558 # apps/api/routers/_init_.py
2559 # -*- coding: utf-8 -*-
2560 from __future__ import annotations
2561 from .health import router as health_router
2562 from .packs import router as packs_router
2563 __all__ = ["health_router", "packs_router"]
2564 # apps/api/routers/health.py
2565 # -*- coding: utf-8 -*-
2566 from __future__ import annotations
2567 from fastapi import APIRouter
2568 router = APIRouter(prefix="/api", tags=["health"])
2569 @router.get("/health")
2570 def health() -> dict[str, str]:
2571     return {"status": "ok"}
2572 # apps/api/routers/packs.py
2573 # -*- coding: utf-8 -*-
2574 from __future__ import annotations
2575 import uuid
2576 from fastapi import APIRouter, Depends
2577 from apps.api.schemas.pack import PackCreate, PackView
2578 from apps.api.services.store import Store
2579 router = APIRouter(prefix="/api", tags=["packs"])
2580 def get_store() -> Store:
2581     project_root = __import__("pathlib").Path(__file__).resolve().parents[3]
2582     data_dir = project_root / "data"
2583     db_path = data_dir / "app.db"
2584     return Store(db_path=db_path)
2585 @router.get("/packs", response_model=list[PackView])
2586 def list_packs(store: Store = Depends(get_store)) -> list[PackView]:
2587     store.seed_demo()
2588     rows = store.list_packs()
2589     return [
2590         PackView(
2591             pack_id=r.pack_id,
2592             pack_code=r.pack_code,
2593             vendor=r.vendor,
2594             series_cells=r.series_cells,
2595             parallel_cells=r.parallel_cells,
2596             rated_capacity_ah=r.rated_capacity_ah,
2597             created_at=r.created_at,
2598         )
2599         for r in rows
2600     ]

```

```

2601 @router.post("/packs", response_model=dict[str, str])
2602 def create_pack(dto: PackCreate, store: Store = Depends(get_store)) ->
2603 dict[str, str]:
2604     pack_id = f"PACK-{uuid.uuid4().hex[:8].upper()}"
2605     store.create_pack(
2606         pack_id=pack_id,
2607         pack_code=dto.pack_code,
2608         vendor=dto.vendor,
2609         series_cells=dto.series_cells,
2610         parallel_cells=dto.parallel_cells,
2611         rated_capacity_ah=dto.rated_capacity_ah,
2612     )
2613     return {"pack_id": pack_id}
2614 # apps/api/domain/__init__.py
2615 # apps/api/domain/enums.py
2616 # -*- coding: utf-8 -*-
2617 from __future__ import annotations
2618 from enum import Enum
2619 class BalanceMode(str, Enum):
2620     passive = "passive"
2621     active = "active"
2622 class AlarmLevel(str, Enum):
2623     info = "info"
2624     warn = "warn"
2625     critical = "critical"
2626 # apps/api/services/__init__.py
2627 # apps/api/services/store.py
2628 # -*- coding: utf-8 -*-
2629 from __future__ import annotations
2630 import sqlite3
2631 from dataclasses import dataclass
2632 from datetime import datetime
2633 from pathlib import Path
2634 def utc_now() -> str:
2635     return datetime.utcnow().strftime("%Y-%m-%d %H:%M:%S")
2636 @dataclass(frozen=True)
2637 class PackRow:
2638     pack_id: str
2639     pack_code: str
2640     vendor: str
2641     series_cells: int
2642     parallel_cells: int
2643     rated_capacity_ah: float
2644     created_at: str
2645 class Store:
2646     def __init__(self, db_path: Path) -> None:
2647         self.db_path = db_path
2648         self.db_path.parent.mkdir(parents=True, exist_ok=True)
2649         self._init_db()
2650     def _connect(self) -> sqlite3.Connection:

```

```
2651         con = sqlite3.connect(str(self.db_path))
2652         con.row_factory = sqlite3.Row
2653         return con
2654     def _init_db(self) -> None:
2655         with self._connect() as con:
2656             con.execute(
2657                 """
2658                 create table if not exists packs(
2659                     pack_id text primary key,
2660                     pack_code text not null,
2661                     vendor text not null,
2662                     series_cells integer not null,
2663                     parallel_cells integer not null,
2664                     rated_capacity_ah real not null,
2665                     created_at text not null
2666                 );
2667                 """
2668             )
2669             con.commit()
2670     def seed_demo(self) -> None:
2671         rows = self.list_packs()
2672         if rows:
2673             return
2674         self.create_pack(
2675             pack_id="PACK-0001",
2676             pack_code="SZ-ESS-0001",
2677             vendor="示例能源",
2678             series_cells=96,
2679             parallel_cells=1,
2680             rated_capacity_ah=280.0,
2681         )
2682         self.create_pack(
2683             pack_id="PACK-0002",
2684             pack_code="SZ-ESS-0002",
2685             vendor="示例能源",
2686             series_cells=96,
2687             parallel_cells=1,
2688             rated_capacity_ah=280.0,
2689         )
2690     def create_pack(
2691         self,
2692         *,
2693         pack_id: str,
2694         pack_code: str,
2695         vendor: str,
2696         series_cells: int,
2697         parallel_cells: int,
2698         rated_capacity_ah: float,
2699     ) -> None:
2700         with self._connect() as con:
```

```

2701         con.execute(
2702             """
2703             insert into packs(pack_id, pack_code, vendor, series_cells,
2704 parallel_cells, rated_capacity_ah, created_at)
2705             values(?,?,?,?,?,?)
2706             """
2707             (pack_id, pack_code, vendor, int(series_cells),
2708 int(parallel_cells), float(rated_capacity_ah), utc_now()),
2709         )
2710         con.commit()
2711     def list_packs(self) -> list[PackRow]:
2712         with self._connect() as con:
2713             cur = con.execute(
2714                 """
2715                 select pack_id, pack_code, vendor, series_cells,
2716 parallel_cells, rated_capacity_ah, created_at
2717                 from packs order by created_at desc
2718                 """
2719             )
2720             return [
2721                 PackRow(
2722                     pack_id=r["pack_id"],
2723                     pack_code=r["pack_code"],
2724                     vendor=r["vendor"],
2725                     series_cells=int(r["series_cells"]),
2726                     parallel_cells=int(r["parallel_cells"]),
2727                     rated_capacity_ah=float(r["rated_capacity_ah"]),
2728                     created_at=str(r["created_at"]),
2729                 )
2730             for r in cur.fetchall()
2731         ]
2732 # apps/__init__.py
2733 # apps/api/__init__.py
2734 # apps/api/schemas/__init__.py
2735 # apps/api/schemas/pack.py
2736 # -*- coding: utf-8 -*-
2737 from __future__ import annotations
2738 from pydantic import BaseModel, Field
2739 class PackCreate(BaseModel):
2740     pack_code: str = Field(..., description="电池包编号")
2741     vendor: str = Field(..., description="厂家")
2742     series_cells: int = Field(..., ge=1, le=400, description="串联数量")
2743     parallel_cells: int = Field(..., ge=1, le=40, description="并联数量")
2744     rated_capacity_ah: float = Field(..., gt=0, description="额定容量(Ah)")
2745 class PackView(BaseModel):
2746     pack_id: str
2747     pack_code: str
2748     vendor: str
2749     series_cells: int
2750     parallel_cells: int

```

```

2751     rated_capacity_ah: float
2752     created_at: str
2753 # apps/api/tests/__init__.py
2754 # apps/api/tests/test_health.py
2755 # -*- coding: utf-8 -*-
2756 from __future__ import annotations
2757 import unittest
2758 class HealthTest(unittest.TestCase):
2759     def test_import_app(self) -> None:
2760         from apps.api.main import app
2761         self.assertIsNotNone(app)
2762     def test_health_endpoint(self) -> None:
2763         from fastapi.testclient import TestClient
2764         from apps.api.main import app
2765         c = TestClient(app)
2766         r = c.get("/api/health")
2767         self.assertEqual(r.status_code, 200)
2768         self.assertEqual(r.json().get("status"), "ok")
2769 if __name__ == "__main__":
2770     unittest.main()
2771 # apps/web/admin.html
2772 <!doctype html>
2773 <html lang="zh-CN"><head><meta charset="utf-8" /><meta name="viewport"
2774 content="width=device-width, initial-scale=1" />
2775 <title>管理员端 - 智能储能电池均衡控制系统</title></head><body style="font-
2776 family: Segoe UI, Microsoft YaHei, sans-serif; padding: 18px;">
2777 <h1>管理员端</h1>
2778 <p>面向管理员：参数模板管理、角色权限、审计留痕与策略版本治理。</p>
2779 <ol>
2780     <li>维护阈值模板（电压差、温升、内阻漂移）。</li>
2781     <li>管理用户与角色授权，配置操作留痕策略。</li>
2782     <li>发布策略版本并维护回滚点。</li>
2783 </ol>
2784 </body></html>
2785 # apps/web/index.html
2786 <!doctype html>
2787 <html lang="zh-CN">
2788 <head>
2789     <meta charset="utf-8" />
2790     <meta name="viewport" content="width=device-width, initial-scale=1" />
2791     <title>智能储能电池均衡控制系统</title>
2792     <style>
2793         :root{--bg:#0b1020;--panel:#111a35;--text:#e9eeff;--muted:#a7b2da;--
2794         accent:#63f2d2;--warn:#ffb44a;--danger:#ff5b6e;}
2795         body{margin:0;font-family: Segoe UI, Microsoft YaHei, sans-serif;
2796         background: linear-gradient(180deg, #070a14, var(--bg)); color: var(--text);}
2797         header{position: sticky; top: 0; backdrop-filter: blur(10px);
2798         background: rgba(17, 26, 53, .78); border-bottom: 1px solid rgba(99, 242, 210, .18);
2799         padding: 14px 18px; display: flex; align-items: center; justify-content: space-
2800         between; gap: 12px;}

```



```
2801     header .brand{font-weight:800;letter-spacing:.5px;}
2802     header .badge{font-size:12px;color:var(--accent);border:1px solid
2803     rgba(99,242,210,.35);padding:4px 10px;border-radius:999px;}
2804     main{max-width:1180px;margin:18px auto;padding:0 18px 28px;display:grid;
2805     grid-template-columns:260px 1fr;gap:16px;}
2806     nav{background:rgba(17,26,53,.72);border:1px solid rgba(99,242,210,.12);
2807     border-radius:14px;padding:12px;}
2808     nav a{display:block;color:var(--muted);text-decoration:none;padding:10px
2809     10px;border-radius:10px;}
2810     nav a:hover{background:rgba(99,242,210,.08);color:var(--text);}
2811     .panel{background:rgba(17,26,53,.72);border:1px solid rgba(99,242,210,
2812     .12);border-radius:14px;padding:14px;}
2813     .k{font-family:ui-monospace,SFMono-Regular,Menlo,Consolas,monospace;
2814     color:var(--accent);}
2815     .hidden{display:none !important;}
2816     .grid{display:grid;grid-template-columns:repeat(3,1fr);gap:12px;}
2817     .card{background:rgba(9,12,24,.55);border:1px solid rgba(99,242,210,.12);
2818     border-radius:12px;padding:12px;}
2819     .card h3{margin:0 0 8px;font-size:14px;color:var(--accent);}
2820     .hint{color:var(--muted);font-size:12px;line-height:1.6;}
2821     .row{display:grid;gap:10px;margin-top:10px;}
2822     input{padding:10px 12px;border-radius:10px;border:1px solid rgba(99,242,
2823     210,.18);background:rgba(255,255,255,.05);color:var(--text);}
2824     button{cursor:pointer;border-radius:10px;border:1px solid rgba(99,242,
2825     210,.22);background:rgba(99,242,210,.10);color:var(--text);padding:8px 12px;}
2826     button:hover{background:rgba(99,242,210,.16);}
2827     button.danger{border-color:rgba(255,91,110,.35);background:rgba(255,91,
2828     110,.12);}
2829     #modal{position:fixed;inset:0;background:rgba(0,0,0,.55);display:flex;
2830     align-items:center;justify-content:center;}
2831     #modal .dialog{width:min(720px,92vw);background:rgba(17,26,53,.96);
2832     border:1px solid rgba(99,242,210,.18);border-radius:14px;box-shadow:0 24px
2833     80px rgba(0,0,0,.45);padding:14px;}
2834     #modal header{position:static;border:none;background:transparent;
2835     padding:0 0 10px;}
2836     #modal .title{font-weight:800;}
2837     </style>
2838 </head>
2839 <body>
2840 <div id="loginView">
2841   <div style="max-width:520px;margin:10vh auto;padding:18px" class="
2842   "panel">
2843     <h1 style="margin:0 0 8px;font-size:18px;
2844     ">智能储能电池均衡控制系统</h1>
2845     <div class="hint">V1.0 演示环境：用于截图与材料生成的最小可运行界面。<
2846     /div>
2847     <form id="login-form" class="row">
2848       <label class="hint">用户名 <input name="username" value="admin"
2849       autocomplete="username" /></label>
2850       <label class="hint">密码 <input name="password" type="password"
```

```
2851 value="admin123" autocomplete="current-password" /></label>
2852     <button type="submit">登录</button>
2853     <div class="hint">提示：任意账号可登录（用于演示与截图）。</div>
2854 </form>
2855 </div>
2856 </div>
2857 <div id="appView" style="display:none;">
2858     <header>
2859         <div class="brand">电池均衡控制台</div>
2860         <div class="badge">V1.0</div>
2861     </header>
2862     <main>
2863         <nav>
2864             <a href="#home">首页</a>
2865             <a href="#packs">电池包</a>
2866             <a href="#cells">单体电芯</a>
2867             <a href="#balancing">均衡策略</a>
2868             <a href="#alarms">告警中心</a>
2869             <a href="#reports">统计报表</a>
2870             <a href="#audit">审计留痕</a>
2871         </nav>
2872         <section class="panel">
2873             <div style="display:flex;align-items:center;justify-content:space-
2874 between;gap:12px;">
2875                 <h2 id="pageTitle" style="margin:0;font-size:16px;">首页</h2>
2876                 <div class="hint">API: <span class="k" id=
2877 "apiStatus">未检测</span></div>
2878             </div>
2879             <div id="pageBody" style="margin-top:10px;"></div>
2880         </section>
2881     </main>
2882 </div>
2883 <div id="modal" class="modal-shell modal hidden">
2884     <div class="dialog" role="dialog" aria-modal="true">
2885         <header>
2886             <div class="title" id="modalTitle">操作</div>
2887             <button class="danger" id="modalClose">关闭</button>
2888         </header>
2889         <div class="hint" id="modalContent"></div>
2890     </div>
2891 </div>
2892 <script>
2893     const modal = document.getElementById('modal');
2894     const modalTitle = document.getElementById('modalTitle');
2895     const modalContent = document.getElementById('modalContent');
2896     function closeModal(){ modal.classList.add('hidden'); }
2897     function openModal(title, content){ modalTitle.textContent=title;
2898 modalContent.textContent=content; modal.classList.remove('hidden'); }
2899     document.getElementById('modalClose').onclick = closeModal;
2900     window.__rkModal = { closeModal, openModal };
```

```

2901     async function apiGet(p){ const r=await fetch(p); if(!r.ok) throw new
2902     Error('HTTP '+r.status); return await r.json(); }
2903     function setStatus(ok){ document.getElementById('apiStatus').textContent
2904     = ok ? 'OK' : 'FAIL'; }
2905     async function ping(){ try{ await apiGet('/api/health'); setStatus(true);
2906     } catch(e){ setStatus(false);} }
2907     function renderHome(){
2908         document.getElementById('pageTitle').textContent='首页';
2909         document.getElementById('pageBody').innerHTML = `
2910             <div class='grid'>
2911                 <div class='card'><h3>均衡目标</h3><div class=
2912 'hint'>以“最小温升、最小内阻偏差、最短均衡时间”为约束，提升电芯一致性。</div>
2913 </div>
2914                 <div class='card'><h3>在线监测</h3><div class=
2915 'hint'>电压/温度/SoC/内阻偏差实时采集，并形成追溯报表。</div></div>
2916                 <div class='card'><h3>闭环留痕</h3><div class=
2917 'hint'>策略下发、执行窗口、告警处置与复核结论自动记录。</div></div>
2918                 </div>`;
2919     }
2920     function renderSimple(title, id){
2921         document.getElementById('pageTitle').textContent=title;
2922         const body = document.getElementById('pageBody');
2923         const descMap = {
2924             zones: '按站点/柜/簇划分资产边界，提供分区统计、权限隔离与设备定位。',
2925             inspections: '对关键页面与接口执行健康检查，形成可追溯的校验记录。',
2926             alerts: '对电压差、温升、采样缺失等风险给出提醒，并引导处置闭环。',
2927             irrigation: '记录策略下发、执行窗口与回滚点，支撑版本治理与审计。',
2928             reports: '输出均衡效果与告警处置报表，支持导出与验收留痕。'
2929         };
2930     };
2931     const desc = descMap[id] || '该模块用于展示业务信息与形成材料截图。';
2932     body.innerHTML = `<div id='${id}' class='card'><h3>${title}</h3><div
2933 class='hint'>${desc}</div><div class='hint'>模块标识：<span class='k'>#${id}
2934 </span></div></div>`;
2935     }
2936     function renderStations(){
2937         document.getElementById('pageTitle').textContent =
2938         '工作台（Stations）';
2939         const body = document.getElementById('pageBody');
2940         body.innerHTML = `
2941             <div id="stations" class="card">
2942                 <h3>Stations 列表</h3>
2943                 <div class="hint">用于演示“新增/编辑/查看/删除”弹窗操作按钮文案。<
2944 /div>
2945                 <div class="toolbar">
2946                     <button onclick="window.__rkModal.openModal('新增',
2947 '创建新的均衡任务并填写字段。')">新增</button>
2948                     <button onclick="window.__rkModal.openModal('编辑',
2949 '修改任务阈值与执行窗口。')">编辑</button>
2950                     <button onclick="window.__rkModal.openModal('查看',

```

```
2951 '查看任务详情与引用关系。')">查看</button>
2952     <button class="danger" onclick="window.__rkModal.openModal('删除
2953 '删除条目前请确认无引用并记录审计。')">删除</button>
2954 </div>
2955     <div class="hint">提示：请按业务流程依次执行新增、编辑、查看、删除
2956 ，并复核弹窗信息完整性。</div>
2957 </div>;
2958 }
2959 function route(){
2960     const hash=(location.hash||'#home').replace(/^#/,"");
2961     if(hash==='home'||hash==='') return renderHome();
2962     if(hash==='zones') return renderSimple('资源分区','zones');
2963     if(hash==='stations') return renderStations();
2964     if(hash==='inspections') return renderSimple('交互检查','inspections');
2965     if(hash==='alerts') return renderSimple('风险提示','alerts');
2966     if(hash==='irrigation') return renderSimple('发布流水','irrigation');
2967     if(hash==='reports') return renderSimple('统计报表','reports');
2968     return renderSimple('页面', hash);
2969 }
2970 document.getElementById('login-form').addEventListener('submit', (e)=>{
2971     e.preventDefault();
2972     localStorage.setItem('__demo_logged_in','1');
2973     document.getElementById('loginView').style.display='none';
2974     document.getElementById('appView').style.display='block';
2975     ping(); route();
2976 });
2977 window.addEventListener('hashchange', ()=>route());
2978 if(localStorage.getItem('__demo_logged_in')==='1'){
2979     document.getElementById('loginView').style.display='none';
2980     document.getElementById('appView').style.display='block';
2981     ping(); route();
2982 } else { renderHome(); }
2983 </script>
2984 </body>
2985 </html>
2986 # apps/web/student.html
2987 <!doctype html>
2988 <html lang="zh-CN"><head><meta charset="utf-8" /><meta name="viewport"
2989 content="width=device-width, initial-scale=1" />
2990 <title>值班员端 - 智能储能电池均衡控制系统</title></head><body style="font-
2991 family:Segoe UI,Microsoft YaHei,sans-serif; padding:18px;">
2992 <h1>值班员端</h1>
2993 <p>面向值班员：告警确认、策略下发、执行窗口控制与风险提示。</p>
2994 <ol>
2995     <li>确认告警并选择处置路径（观察/降载/停用）。</li>
2996     <li>查看均衡策略建议与预计温升。</li>
2997     <li>一键生成处置报告并提交复核。</li>
2998 </ol>
2999 </body></html>
3000 # apps/web/teacher.html
```